

RESEARCH ARTICLE



WILEY

The favela effect: Spatial inequalities and firm strategies in disadvantaged urban communities

Leandro S. Pongeluppe 

Rotman School of Management,
University of Toronto, Toronto, Ontario,
Canada

Correspondence

Leandro S. Pongeluppe, Rotman School
of Management, University of Toronto,
Toronto, ON M5S 3E6, Canada.
Email: l.pongeluppe@rotman.utoronto.ca

Abstract

Research summary: E-commerce firms make fewer products available and charge higher delivery prices to customers inside Brazilian favelas than they do to customers immediately outside favelas, despite the absence of infrastructure impediments at the favela borders. This phenomenological study uses mixed methods to investigate firm heterogeneity in these practices. The analysis shows that some firms treat favela consumers more equitably than their competitors. These firms (i) invest in physical stores inside and outside favelas, which are complementary to their online marketplaces, and (ii) engage genuinely with employees and consumers, which reflects their stakeholder orientation. By examining how firms operate in disadvantaged communities, scholars can enrich core theoretical constructs in strategic management, particularly by integrating insights from the fields of critical geography and urban economics.

Managerial summary: This study investigates whether firms operate differently in disadvantaged communities compared to co-located nondisadvantaged areas. Findings show that operations in disadvantaged communities, such as favelas (Brazilian urban slums), demand specific investments that support transactions and contribute to realizing the underdeveloped potential of those communities. Firms succeed in commercial endeavors within disadvantaged communities by redeploying their resources and cultivating a stakeholder culture concomitantly. This strategy

enables superior performance and the change-making of structural inequalities to help alleviate poverty and develop urban communities.

KEYWORDS

complementary assets, disadvantaged communities, mixed methods, socioeconomic inequality, spatial inequality, stakeholder theory, sustainable development goals

“So, there are some communities [favelas] where nothing bought from the internet gets delivered. Nobody gets anything, and nothing arrives there. (...) The person is blocked during the acquisition by the system that prevents the purchase based on the zip code input. So, you have communities where nothing gets delivered.”—
Social Firm Manager at Favela da Rocinha (Rio de Janeiro).

1 | INTRODUCTION

Research in strategic management is moving toward a new paradigm that aims to understand problems of profound societal importance (George, Howard-Grenville, Joshi, & Tihanyi, 2016; McGahan, 2018; Tihanyi, 2020). Understanding the nuanced ways in which firms engage with consumers in disadvantaged communities is an important opportunity aligned with this goal. Remediation of poverty is compelling morally, socially, economically, and for the sustainability of firms that seek to engage with these communities. Brazilian favelas (urban slums) are the archetypical disadvantaged communities as they are characterized by low-income households, the absence of well-defined property rights, mediocre law enforcement systems, less developed capital and labor markets, and less developed health and education systems (Marx, Stoker, & Suri, 2013; Meirelles & Athayde, 2014; Perlman, 2010). This article addresses the question: Do firms operate in disadvantaged communities differently from closely located nondisadvantaged areas? In doing so, the study emphasizes opportunities for scholars of strategic management to incorporate insights from other fields—in particular, critical geography and urban economics—to improve our understanding of how firm behavior, especially in disadvantaged areas, may contribute to the persistence of disenfranchisement.

Favela boundaries are both physically open and sharply distinct. They are physically open in the sense that there are no gates, walls, or other infrastructure impediments that block the entry or exit of favelas. The roads into Brazilian favelas are open to an entering traveler who moves from outside the favela across the favela border, and similarly to an exiting traveler who moves from inside the favela into the surrounding neighborhood. Yet favela boundaries are sharply distinct because their housing stock and architectural features are immediate and dramatically dissimilar from the surrounding areas. This article identifies this paradoxical boundary as a “spatial inequality.” Following critical geography studies (Davis, 2006; Melgaço & Prouse, 2017; Santos, 1978, 1987, 2017; Zenou & Boccard, 2000), geographical spaces (e.g., places) are the result of power dynamics in society, which socially construct these spaces, imbue meaning to their boundaries, and limit the socioeconomic opportunities of their inhabitants. Thus, “spatial inequalities” are defined as the geographical and physical differences



FIGURE 1 Spatial inequality between *Gávea* Neighborhood and *Favela da Rocinha* (Rio de Janeiro, Brazil). Image credit: Johnny Miller, Unequal Scenes. Additional illustrations of this global phenomenon can be found on the “Unequal Scenes” project website. Available at: <https://unequalscenes.com/>. Accessed on March 07th, 2022

between areas that are both a result and driver of socioeconomic processes.¹ This definition implies that physical space is not a mere background to economic activity. Instead, it has an active role in shaping, expressing, and reinforcing the socioeconomic structures of production, consumption, and oppression embedded in society. Figure 1 presents a visual illustration of “spatial inequality” at a favela boundary.

The study follows a phenomenological approach (Flammer & Ioannou, 2021; Hambrick, 2007; Helfat, 2007) to report and document facts that are understudied and underdeveloped in the strategic management literature. Mixed methods that include econometric models and ethnographic techniques are employed (Kaplan, 2016; Small, 2011). The econometric analysis relies on a spatial regression discontinuity design (RDD) (Campa & Serafinelli, 2019; Flammer, 2015; Flammer & Bansal, 2017) to determine how e-commerce firms treat different types of consumers. The qualitative analysis consists of participant observations and interviews (Kaplan, 2008; Pratt & Kim, 2012). The findings from the qualitative component

¹Often, economic geographers consider income inequality as the main feature of spatial inequalities. In this article, I focus on the implications of spatial inequality for income inequality, and thus I define the term more narrowly than some economic geographers have.

lend support to the main regression results and suggest mechanisms and patterns behind the discovered heterogeneity in firm strategies toward disadvantaged communities.

The results confirm that, when e-commerce firms are aware that a customer lives inside a favela, they are more likely to refuse orders and/or charge higher delivery prices compared to outside favelas. Moreover, the results show that customers located further inside a favela are charged much higher delivery prices than those at the border. On average, a customer located 500 m (about 0.31 miles) inside a favela is charged a 45% higher delivery price than a customer living just inside the favela border. Interestingly, there is heterogeneity across firms, with some firms charging favela customers significantly less than their competitors.

These results carry important implications for the strategic management literature. In line with the literature on critical geography and urban economics (Chown & Liu, 2015; Chyn, 2018; Davis, 2006; Lucci, Bhatkal, & Khan, 2018; Marx et al., 2013; Melgaço & Prouse, 2017; Santos, 1978, 1987, 2017; Zenou & Boccoard, 2000), the study shows that spatial inequality, which often distinguishes marginalized and economically disadvantaged populations, reinforces socioeconomic disparities. I refer to this phenomenon as “the favela effect.” This is because firms charge consumers in disadvantaged communities higher prices and serve them with different product assortments. In doing so, firms use the information on individuals’ spatial locations to limit their access to goods and services.

Some firms, however, develop strategies to overcome spatial inequalities and deal with disadvantaged communities more equitably than their competitors (Bansal & DesJardine, 2014; Dorobantu & Odziemkowska, 2017; Dutt et al., 2016; Flammer & Bansal, 2017; Flammer & Luo, 2017; George et al., 2016; George, McGahan, & Prabhu, 2012). Such firms accomplish this by enhancing their trustworthiness within communities (McEvily, Perrone, & Zaheer, 2003) and by conveying their dual orientation (Battilana & Dorado, 2010). In practice, these firms invest in physical stores that support transactions within disadvantaged communities (Barney, 1991; Tripsas, 1997) and reinforce a culture of care toward their stakeholders (Bertels, Howard-Grenville, & Pek, 2016; Eccles, Ioannou, & Serafeim, 2014; Flammer & Kacperczyk, 2016; Flammer & Luo, 2017; Henisz, Dorobantu, & Nartey, 2014; Jones, 1995; Nardi, 2021; Pless & Maak, 2004). As a result, these firms mitigate socioeconomic disparities by engaging in commercial activities with disenfranchised individuals that build over time.

The article contributes to the strategic management literature in several ways. First, the article contributes to theory by departing from a phenomenological approach and integrating concepts and insights from the fields of critical geography and urban economic studies. Second, the research delivers a methodological contribution in showing how mixed methods that combine quasi-causal estimations with on-the-ground interviews and observations might support the development of knowledge about resource-scarce locations. Finally, in showing how firms integrate their resources with a stakeholder orientation to enfranchise disadvantaged communities, the study demonstrates the importance of management in achieving some of the United Nations’ Sustainable Development Goals (SDGs).

2 | WORLD SLUMS AND BRAZILIAN FAVELAS

Slums are prevalent spaces in urban areas across the globe. They are characterized “by overcrowding, poor or informal housing, inadequate access to safe water and sanitation, and insecurity of tenure” (Davis, 2006, p. 23). A recent publication from the United Nations Habitat reports that approximately a billion people live in slums worldwide (UN-Habitat, 2016).

Estimates suggest that, by 2050, 2.3 billion people (or 22.5% of the world population) will live in slum households.² These figures may be underestimated due to differences in the definition of “slums” across countries and difficulties in data collection; the number of slum residents worldwide might be more accurately projected to reach 3 billion people by 2050 (Lucci et al., 2018).

The urban economics literature shows that slums function as “poverty traps” (Dasgupta, 1997; Marx et al., 2013) by locking dwellers in a vicious cycle of deprivation and marginalization across generations (Chyn, 2018; Marx et al., 2013). This problem is particularly acute in the Global South. Critical geography studies show that, since 1970, the slum growth rate surpassed the urbanization growth rate, particularly in developing countries (Davis, 2006, pp. 16–17). This trend, labeled “overurbanization” or “favelization,” amplifies urban poverty and inequality over time and across space and has led to the rise of “megaslums”—conurbations of slums in which vast and continuous areas of informal housing and informal economic activity arise within urban spaces (Davis, 2006, p. 26). Famous megaslums include Dharavi in India, Neza-Chalco-Itza in Mexico, Soweto in South Africa, and Rocinha in Brazil. The slum phenomenon, however, is not limited to developing countries. Residential segregation, redlining policies, South–North migrations, refugee crises, internal inequality, and continuous impoverishment of indigenous populations have increased the prevalence of urban slums in wealthier nations. For example, in South Korea and the United States, respectively, 14.2 and 12.8 million inhabitants live in urban slums (Davis, 2006, p. 24), while France and Spain report an increase in the number of people living in the so-called *bidonvilles*³ and *chabolas*,⁴ respectively.

In Brazil, urban slums are called “*favelas*.” The term dates back to 1890 and originally referred to a type of plant.⁵ As a sociological phenomenon, favelas can be traced to the structure of the Brazilian colonial system. In this system, the relationship between slaves and masters was marked by separation, mutual dependence, and miscegenation, with individuals in each group interacting and cohabitating in sharply delineated but closely proximate spaces in rural and urban contexts (Freyre, 1934, 1936). Social disparity, geographic proximity, and recurring interactions led the favela boundaries to be drawn starkly while remaining open. Other factors that influenced the prevalence of favelas include the emancipation of slaves, the immigration of poor Europeans to South America, internal rural–urban migrations, and rising income inequality in the country (Meirelles & Athayde, 2014; Perlman, 2010).

Especially during the 1970s and 1980s, favelas became the strongholds of drug lords and police conflicts. These problems cultivated civil rights abuses, discrimination, stigmatization, marginalization, and criminalization in favelas (Larkins, 2015). Although these issues have been partially addressed in many communities, for example, with the advent of “pacification” programs such as in Rio de Janeiro favelas prior to the 2014 FIFA World Cup and 2016 Summer Olympic Games (Bellego & Drouard, 2019; Larkins, 2015; Willis & Prado, 2014), they remain

²Our World in Data webpage. Available from <https://ourworldindata.org/urbanization#urban-slum-populations>. Accessed on June 10th, 2021.

³Le Figaro, June 26th, 2019. *En France, près de 500 bidonvilles à résorber*. Available from <http://www.lefigaro.fr/actualite-france/en-france-pres-de-500-bidonvilles-a-resorber-20190626>. Accessed on June 26th, 2019.

⁴El País, June 11th, 2020. *Cada sociedad se retrata en su espacio público*. Available from https://elpais.com/elpais/2020/03/03/seres_urbanos/1583229795_029032.html. Accessed on June 16th, 2020.

⁵Around 1890, veterans of the Brazilian army returned to Rio de Janeiro from a civil war (the *Canudos* war), which took place in the northeastern Brazilian state of Bahia. Despite winning the war, the veterans were not allocated pieces of land as the government had promised before the conflict. As a result, this mass of soldiers occupied the current *Providência* hill close to the Brazilian army headquarters, and named the occupation as “*favela*” (da Cunha, 1905; Meirelles & Athayde, 2014).

identified with favelas and favela dwellers (Jovchelovitch & Priego-Hernandez, 2013; Larkins, 2015; Meirelles & Athayde, 2014).⁶

According to the 2010 Brazilian census, approximately 11.4 million people (about 6% of Brazil's population) live in favelas, but this figure is likely underestimated more than three-fold as the most recent United Nations' estimates dated in 2014 identify 38.5 million people living in Brazilian slums (about 22.3% of Brazil's population).⁷

Figure 2 shows a socioeconomic comparison of the 21 favelas considered in the present study with the areas outside their borders.⁸ The panels show that favelas perform significantly worse than their immediate surroundings on all indicators. When favelas are compared to metropolitan areas and the states in which they are situated, they perform worse on at least four socioeconomic indicators, namely child mortality, population density, average income per capita, and human development (measured by the human development index, or HDI). This means that, despite the prime location of favelas in urban centers, their socioeconomic development lags behind even that of remote rural areas. The gap in the HDI between favela dwellers and their immediate surrounding neighbors is analogous to the gap between Sri Lanka and Singapore; or between the Dominican Republic and the United Kingdom (UNDP, 2010, pp. 152–153).⁹

3 | FIRM STRATEGIES IN FAVELAS

Despite socioeconomic disparities, consumption patterns of favela dwellers bear some resemblance to those of their surrounding counterparts in the sense that both favela dwellers and their neighbors analogously consume similar portfolios of basic consumption goods, although wealthier neighbors also consume luxury goods that are unaffordable to favela dwellers (de Duarte, Macau, Flores e Silva, & Sanches, 2019; Meirelles & Athayde, 2014; Perlman, 2010; Villas Bôas, 2020). Anthropological studies indicate that “[r]egardless of household income, a favela family cannot buy acceptance, legitimacy, or title to their land, so they invest in home improvements, domestic appliances, and personal status symbols” (Perlman, 2010, pp. 255–256). According to Perlman (2010), a favela household in 2003 had similar domestic appliances as non-favela residences in Rio de Janeiro, despite potential differences in appliances brands, prices, quality, and usage intensity within the household. Specifically, 97% of favela households had a television, 97% a refrigerator, 94% a stereo system, and so on. Meirelles and Athayde (2014) show that, in 2013, half of favela households had a physical connection to the internet, and 85% of favela residents used smartphones and had a Facebook® account (Meirelles & Athayde, 2014, pp. 93–94). Villas Bôas (2020) indicates that favela consumers purchase products through comparable channels to their

⁶Several studies evaluate the likelihood of individuals exiting favelas, given the blatant discrimination suffered by these populations. While some studies show that favelas operate as “poverty straps” (Marx et al., 2013, p. 190), locking individuals into these communities; other studies show that favela populations, particularly the ones with slightly higher education levels, tend to move out in the long-run (Perlman, 2010, p. 233). Finally, some studies describe that many dwellers deeply identify with their communities (Jovchelovitch & Priego-Hernandez, 2013) and enjoy living in the favela despite the difficulties. These individuals often do not want to move out even if they have a chance (Meirelles & Athayde, 2014).

⁷Millennium Development Goals Indicators for Brazil. Available from <http://mdgs.un.org/unsd/mdg/Data.aspx?cr=76>.

⁸Supporting Information Appendix 1 presents more details on the comparison between favelas and surroundings in the sample.

⁹Based on the Brazilian 2010 Census data, the average 2010 HDI in favelas was 0.662 (Medium Human Development level), while in the surroundings was 0.843 (Very High Human Development level). Respectively, these values are similar to 2010 HDI levels in countries such as Sri Lanka (0.658) or the Dominican Republic (0.663) for favelas; and Singapore (0.846) or the United Kingdom (0.849) for the surroundings (UNDP, 2010, pp. 152–153).

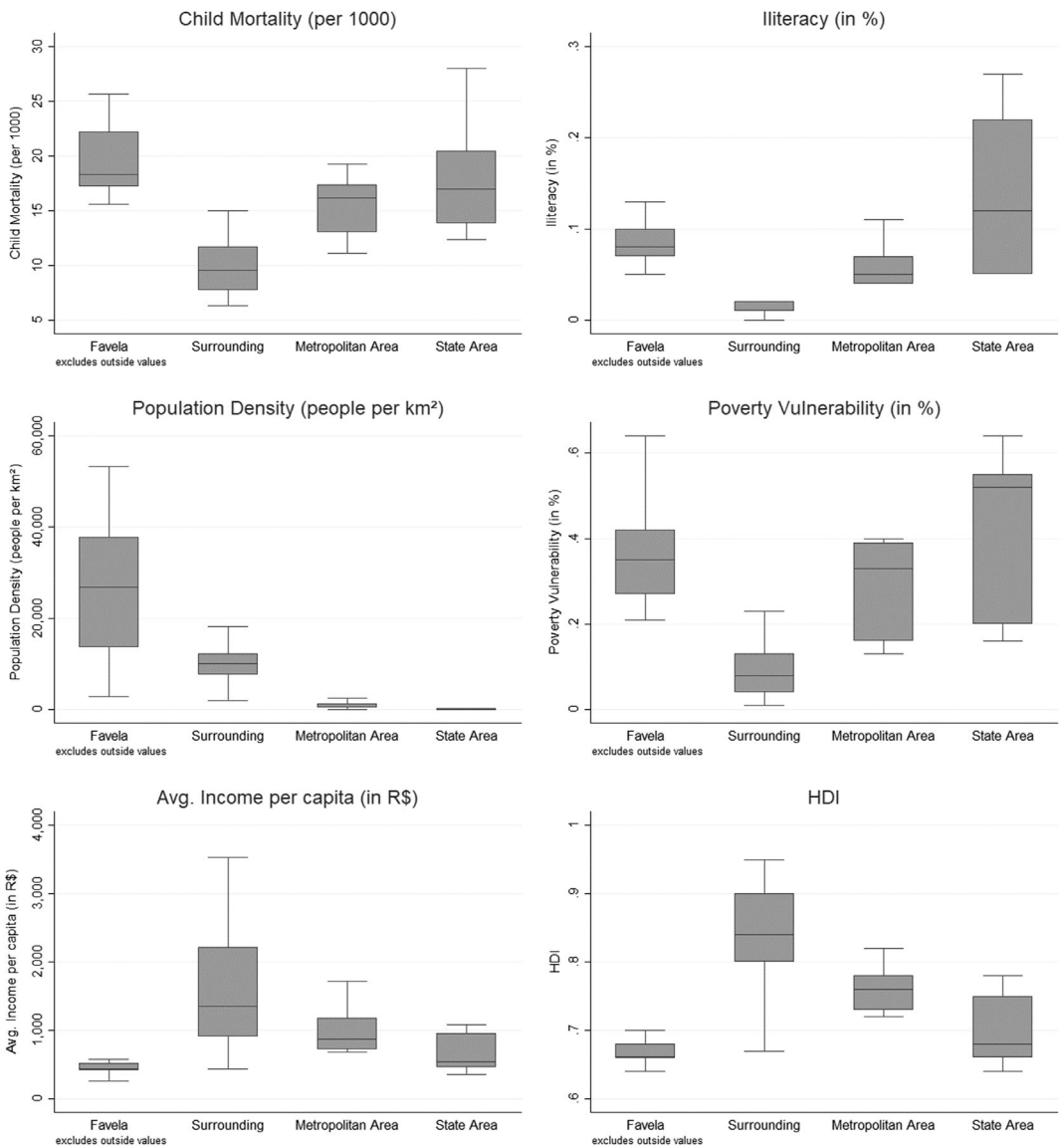


FIGURE 2 Socioeconomic indicators comparison among favelas, surroundings, metropolitan areas, and state areas. *Source:* Brazilian National 2010 Census Data collected at the *Atlas do Desenvolvimento Humano no Brasil*. Data on the 21 favela-surrounding dyads present in the sample as well as comparison with their respective metropolitan areas and state areas (urban and rural areas). Note that the y-axes of the graphs are on different scales

surrounding counterparts, i.e. both in physical and online marketplaces. The same report estimates that favela consumers have an aggregate purchasing power of R\$ 119.8 billion Brazilian reais (\$ 28.7 billion US dollars¹⁰, for more information, see Supporting Information Appendix 13).

¹⁰In this conversion, I consider that US\$ 1.00 equal to R\$ 4.18, which was the exchange rate on January 24th, 2020, the date in which the report was published.

Favela residents do not receive the same level of service as their surrounding counterparts when performing online transactions, as described in the opening quote of the article. The quote indicates that some e-commerce firms make no products available inside the favela. The issue of nondelivery arises from specific firm behavior rather than from theft risk inside the favela (for more information, see Supporting Information Appendix 11). Moreover, the informant suggested that e-commerce firms refused to transact with favelas consumers despite their purchasing power. Zip code information leads favela consumers to be screened-out during the transaction process. When e-commerce firms obtained information during the transaction about consumers' locations, they adjusted their pricing strategies and product assortments accordingly (Acquisti, Brandimarte, & Loewenstein, 2015; Shiller, 2014).

This screening process exemplifies the findings of critical geography studies indicating that spatial inequality (reflected in zip codes) perpetuates socioeconomic disparity in access to goods, services, and opportunities. After accessing information about consumers' locations, e-commerce firms used several tactics to avoid transacting with favela consumers. First, they refused to accept orders as described by the informant. Second, they may increase the delivery prices as a mechanism for discouraging the finalization of the purchase.

Similar evidence from the United States shows that Amazon has been accused of delivery discrimination toward African-American and Hispanic communities by refusing to offer same-day delivery in these neighborhoods (Galhotra, Brun, & Meliou, 2017; Ingold & Soper, 2016).¹¹ Comparable findings have been reported in studies about "food deserts" (Allcott et al., 2019; Beaulac, Kristjansson, & Cummins, 2009; Bitler & Haider, 2011; Gordon et al., 2011). According to this literature, disenfranchised individuals face a double penalty when trying to access products and services precisely because of the geographical location of their residences.

4 | DATA AND METHODS

This study exploits a large and unique dataset describing 406,900 transaction quotes by 11 e-commerce firms operating in Brazil, including both local and international firms.¹² The dataset describes 13 distinct products offered by each of the firms, with the products classified into four categories according to their value (cheap vs. expensive) and size (small vs. large).¹³ Data collection was performed through a bot written in Java[®]. The bot entered each e-commerce website and, based on a preliminary selection of the products, simulated a purchase with a delivery location at a specific zip code. Prices were obtained on each product by each firm in 282 different zip codes, 48% of which were inside one of twenty-one different favelas—on the main

¹¹USAToday. April 22nd, 2016. *Amazon same-day delivery less likely in black areas, report says*. Available from <https://www.usatoday.com/story/tech/news/2016/04/22/amazon-same-day-delivery-less-likely-black-areas-report-says/83345684/>. Accessed on October 24th, 2019. In response, Amazon argued that low demand and logistical difficulties explained the absence of same-day delivery. After the issuance of the report, Amazon changed its approach.

¹²The sample of firms was defined based on the Brazilian Society of Retail and Consumption (pt. *Sociedade Brasileira de Varejo e Consumo*, SBVC) ranking of the biggest and most popular e-commerce firms operating in the country in 2016 (SBVC, 2016). I considered three criteria to include a firm in the sample: (i) whether the firm was among the biggest/most popular e-commerce firms according to the SBVC; (ii) whether the firm had national reach; and (iii) whether the firm offered a wide range of products. Supporting Information Appendix 6 presents more information about each firm.

¹³The choice of these categories aims to control for product-specific characteristics that affect delivery strategies.

Therefore, heterogeneity on product value and size does not bias the main results. Also, this broad range of products allowed testing for product heterogeneity effect (see Supporting Information Appendix 2).

roads to avoid logistical hurdle biases—and 52% of which were immediately outside the favelas.¹⁴ Subsequently, the bot scraped for each product, firm, and zip code the following information: (i) the price of the product at the moment of the transaction quote, (ii) whether or not the firm refused the order after obtaining the zip code, as well the justification for the refusal, (iii) the delivery price, and (iv) the delivery time. Data collection occurred weekly from October 2017 to February 2018, which allows for assessing consistency over time.

The empirical estimation uses a set of logistic (LOGIT) and ordinary least squares (OLS) regression models in a spatial RDD frame (Campa & Serafinelli, 2019; Flammer, 2015; Flammer & Bansal, 2017; Imbens & Lemieux, 2008; Lee & Lemieux, 2010; Moore, 2009). The analysis considers three main dependent variables Y_{jkzt} , respectively: (i) *Refused order* (dummy), which takes the value 1 if the e-commerce firm refused the transaction and 0 if accepted; (ii) *Delivery price* (in Brazilian Reais—R\$); and (iii) *Delivery price ratio* (in %), that is, the delivery price over the total price of the product; for each e-commerce firm j , for each product category k , for each zip code z , for each week t . The independent variable used to capture the study's main effect is a “Favela” (dummy), which takes the value 1 for zip codes inside and 0 for zip codes outside each one of the twenty-one favelas. The model included a set of fixed effects Z_{jkzt} to represent the firm, product category, location, and time (week).^{15,16} Lastly, the models have clustered standard errors at the 21 favela-surrounding dyads to avoid misleadingly small standard errors (Angrist & Pischke, 2008; Cameron & Miller, 2015; Cameron & Trivedi, 2010).¹⁷ The main regression model is the following:

$$Y_{jkzt} = \beta_0 + \beta_1 \text{Favela}_{jkzt} + \beta_2' Z_{jkzt} + e_{jkzt} \quad (1)$$

¹⁴I selected the 21 favelas based on their size and geographical location in Brazil. Following the geographical distribution of favelas described by the Brazilian 2010 Census, I included all the biggest favelas of all the major state capitals in the country to avoid possible regional biases. Nevertheless, the two biggest cities in the country, São Paulo and Rio de Janeiro, have in total 10 favelas included so as to avoid misrepresentation bias of these locations in the final sample. Also, three favelas are included for city of Belém because it is the state capital with the highest percentage of inhabitants living in favelas (54.5% of the total city population). More information for each favela, surroundings, metropolitan area, and state area (urban and rural) can be found in the Supporting Information Appendix 1. The initial database had 385 zip codes, but some of these zip codes were “not found” by Google Earth® when I performed the distance data collection. To be conservative, I removed these zip codes from the sample. Moreover, to ensure that each designated zip code was truly inside the favela, I performed approximately 20 calls to phone numbers associated with the zip codes in Google Maps® software. The purpose was to double-check the accuracy of the zip code list. To this end, I randomly selected phone numbers inside five different favelas from the sample. I made the calls using Skype® during May 2017, before data collection started. To increase the likelihood of answers to the phone calls, I targeted phone numbers of commercial establishments. All the calls confirmed that the zip code information was correct, i.e. the location was indeed inside the designated favela.

¹⁵I ran the same analysis including the distance to the favela border variable and/or with zip code fixed effects, and the results hold.

¹⁶Note that I do not test for serial correlation because the database is not in a panel data format. Despite having the same 11 firms over the 18 weeks, the basket of products offered by each firm can vary weekly depending on a wide range of factors, including prior sales, seasonal demand, event-specific demand, supply availability, and vendor offers. As a result, I cannot compare the very same product offered by each firm over the entire 18 weeks. That is why I opted for crosssectional models with fixed effects.

¹⁷I cluster the standard errors at the favela-surrounding dyads to focus on the comparison between these locations. Moreover, the number of 21 clusters is adequate as the dataset has unbalanced clusters within each dyad (some clusters have more observations than others, ranging from 2.855 to 28.609; Cameron & Miller, 2015). Nevertheless, I ran the same analysis with “robust” standard errors, and the results hold.

The dataset also describes the distance to the favela border of each customer. This variable represents the distance between each zip code and the border of the respective favela (in kilometers). Negative values represent distances outside favelas to the closest border, whereas positive values represent distances from inside favelas to the nearest border.¹⁸ These distances were collected using Google Earth® software. Distance data was collected manually to support a visual accounting of the location of the favela's main entrances when defining the border point.¹⁹ This procedure helped to avoid measurement errors. Supporting Information Appendix 14 shows an example of distance data collection.

Following RDD assumptions, I argue that, given the geographical proximity of the zip codes inside and outside the favelas, both the criminality risk and the logistical cost to transport products from firm warehouses to zip codes inside and outside favelas are similar. However, concerns about the suitability of RDD might arise if crime rates and infrastructure conditions were significantly different at the border.

Additional tests evaluating criminality rates within and outside a subset of favelas in Rio de Janeiro, for which data is available, reduce these concerns to some extent. The results show that there is no systematic difference in the likelihood of crimes—*street robbery* (count), *automobile robbery* (count), *cargo robbery* (count), *store robbery* (count), *home robbery* (count), among others—occurring inside favela relative to the entirety of the Rio de Janeiro Metropolitan area (see Supporting Information Appendix 11).²⁰

I implement a test showing that there is no jump in the discontinuity after controlling for variation in the physical infrastructure of zip codes that lie marginally inside versus marginally outside the favela boundaries (Flammer, 2015; Flammer & Bansal, 2017; Imbens & Lemieux, 2008; Lee & Lemieux, 2010; Moore, 2009).²¹ This test uses information on three infrastructure variables: *electricity coverage* (in %), *garbage collection coverage* (in %), and (iii) *water and sewage coverage* (in %) for each zip code in the dataset (see Supporting Information Appendix 10). Note that the garbage collection measure is especially informative regarding the logistical quality of the terrain because this service requires the transit of trucks and large vehicles.²²

5 | MAIN EFFECT: RESULTS AND DISCUSSION

5.1 | Descriptive statistics

This section provides initial analyses of the data. First, 52,823 inquiries by customers located both inside and outside favelas received the message “refused order.” Of these, 44,004 generated

¹⁸I manually collected the distance data (see illustration in Supporting Information Appendix 14). In this procedure, I considered two factors: (a) the location of the main entry points and (b) the closest possible main entry point to each zip code. Finally, to operationalize the econometric models, distances d were registered with negative values for zip codes outside favelas $d_z^{\text{Surrounding}} < 0$ and with positive values for zip codes inside favelas $d_z^{\text{Favela}} > 0$.

¹⁹Supporting Information Appendix 7 shows a picture to illustrate the entry point in *Favela da Rocinha* (Rio de Janeiro).

²⁰Results might be subject to some degree of reporting biases. As favelas dwellers often do not have insurance against theft, they might be less likely to report some of the crimes described.

²¹This suggests that the logistical challenges of delivering products to zip codes located marginally inside vs. marginally outside the favelas border are roughly equivalent, i.e. the RDD assumption is valid.

²²To provide a vivid illustration of the RDD assumption test on logistical infrastructure inside favelas, check the Washington Post's 360° video report available from https://www.youtube.com/watch?v=x_35zVvJtAg&feature=youtu.be. Note on 0'59' minute a garbage collection truck in operation inside the favela. Accessed on September 13th, 2020.

TABLE 1 Descriptive statistics

#	Variable	Obs.	Min	Max	Mean	SD	Correlation matrix			
							1	2	3	4
1	1 (Favela)	406,900	0	1	0.48	0.50	1.000			
2	Distance to the favela border (km)	406,900	−2.68	1.27	−0.06	0.46	.750	1.000		
3	1 (Refused order)	406,900	0	1	0.12	0.33	.011	.003	1.000	
4	Delivery price (in R\$)	354,077	0.00	1,871.16	86.64	171.42	.042	.072		1.000
5	Delivery price ratio (in %)	354,077	0.00	4.99	0.24	0.33	.041	.077		.360

TABLE 2 Group comparison surroundings and favelas

Transaction aspects	The average on the surroundings	The average on the favelas	Mean difference (favela – surround)	p-value
1 (Refused order)	0.120	0.128	0.008	.000
Delivery price (in R\$)	79.86	94.12	14.26	.000
Delivery price ratio (in %)	0.229	0.256	0.027	.000

Note: Table compares the means inside and outside favelas in the 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on 10 different products of four different categories (by size and value), for 11 different e-commerce firms, operating in the main state capitals of Brazil.

the explanation that the product was out of stock.²³ In addition to these “refused order” occurrences, the bot collected the delivery price and time for 354,077 approved transactions. Accepted transactions accounted for almost 90% of the total sample. In the analysis, all delivery price and delivery price ratios are calculated on orders that were accepted by the firm. Table 1 presents the descriptive statistics of the main variables in the study database.²⁴

A difference-of-means test is presented in Table 2. This assessment shows that the values of all variables are statistically different in favelas and surrounding areas (p -values = .000). On average, e-commerce firms are more likely to refuse an order and charge higher delivery prices to consumers located inside favelas than outside of them. Favela consumers are 0.8 percentage points (p.p.) more likely to have an order refused by the e-commerce firms than consumers outside favelas. Favela customers pay on average R\$ 14.26 Brazilian reais (\$ 4.3 US dollars) more than outside customers for their purchases.²⁵ Similarly, favela dwellers pay a 2.7 p.p. higher delivery price ratio than outside neighbors. These results are consistent with the stipulation that firms screen zip codes to identify whether orders originate inside favelas and adjust prices accordingly.

²³These unavailable transactions may have happened because sometimes the product went out of stock from the moment the bot started scraping data to the point it ended. Alternatively, they might occur if the e-commerce firm decides to decline the transaction after screening the zip code of the consumer.

²⁴It is important to note that refused orders can bias the delivery price strategy given that firms can “cherry-pick” the zip codes they want to transact with *ex ante*. Therefore, I performed a Heckman two steps model (Heckman, 1979; Puhani, 2000) to check the accuracy of the estimators. This model is further described in Supporting Information Appendix 4.

²⁵Throughout the article, I consider the US\$ 1.00 equal to R\$ 3.33, which was the exchange rate on December 31st, 2018.

5.2 | The “favela effect” in online transactions and statistical heterogeneity among firms

The study includes regressions using logistic estimations to assess the likelihood that an e-commerce firm refuses orders that would require delivery to favelas. Table 3 shows this specification. The full controlled LOGIT model (2) shows that, on average, consumers located inside favelas are 6.67% more likely to receive a “refused order” message than their surrounding counterparts ($SE = 0.00191$, $p\text{-value} = .000$).²⁶ In other words, favela consumers are more likely to have their orders refused after e-commerce firms obtain information about the delivery zip code. This statistical difference suggests that e-commerce firms may report to some customers in favelas that an item is “out of stock” when, in fact, it is not.

The analysis includes assessments of the level of delivery charges as well as the percentage of the delivery charge over the price of the product, conditional on e-commerce firms accepting orders. In Table 3, the OLS model (2) shows that, on average, the delivery price to consumers inside favelas is R\$ 18.65 Brazilian reais (\$ 5.6 US dollars) more expensive than the delivery price to consumers outside favelas ($SE = 0.0937$, $p\text{-value} = .000$). This value corresponds to 1.8% of the monthly income of the average favela resident. Table 3 shows the results of an OLS model (4) indicating that, on average, the delivery price ratio inside favelas is 3.06 p.p. higher than outside ($SE = 0.000169$, $p\text{-value} = .000$).

The results are confirmed in a spatial discontinuity analysis on orders immediately inside and outside favelas (Angrist & Pischke, 2008; Campa & Serafinelli, 2019; Flammer, 2015; Flammer & Bansal, 2017; Hahn, Todd, & Van der Klaauw, 2001; Imbens & Lemieux, 2008;

TABLE 3 “The favela effect”

	LOGIT (1)	LOGIT (2)	OLS (1)	OLS (2)	OLS (3)	OLS (4)
	1(Refused order)		Delivery price (in R\$)		Delivery price ratio (in %)	
Favela	0.0665 (0.0435)	0.145 (0.00191)	14.25 (5.390)	18.65 (0.0937)	0.0272 (0.0109)	0.0306 (0.000169)
Week Fixed Effect	N	Y	N	Y	N	Y
Product Category Fixed Effect	N	Y	N	Y	N	Y
Firm Fixed Effect	N	Y	N	Y	N	Y
Location Fixed Effect	N	Y	N	Y	N	Y
Constant	−1.989 (0.0689)	0.513 (0.114)	79.86 (11.93)	19.17 (9.001)	0.229 (0.0251)	0.0610 (0.0184)
Observations	406,900	288,679	354,077	354,077	354,077	354,077
Pseudo and R-squared	0.000	0.147	0.002	0.334	0.002	0.385
Log-Likelihood	−152,318	−114,032				

Note: Standard errors are clustered at 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on 10 different products of four different categories (by size and value), for 11 different e-commerce firms, operating in the main state capitals of Brazil.

²⁶I calculate the probabilities plugging coefficients in the LOGIT formula.

Lee & Lemieux, 2010). Figure 3 shows both a parametric and a nonparametric estimation through a full controlled binned scatter plot (Starr & Goldfarb, 2020).

Most of the “refused orders” occur in zip codes close to the favela border, where there is a considerable discontinuity in the likelihood of refusal. Orders from further inside a favela (positive values in the x -axes) are associated with higher absolute and proportionate delivery charges.

Despite the absence of a discontinuity in the delivery price graphs, the analysis yields evidence of a statistical difference in slope (p -values = .000), also known as a “regression kink” (Nielsen, Sørensen, & Taber, 2010).²⁷ The graphs show that orders with delivery locations 500 m (about 0.31 miles) inside a favela are associated with a delivery charge 45% greater than that at the favela border. For deliveries 1,000 m (about 0.62 miles) inside the favela, the delivery charge is 90% greater than for deliveries at the border. Each additional meter (a stride distance) between the border and the delivery location within the favela center is associated with a 0.1% higher delivery charge.²⁸ Firms operate in a two-step model (Heckman, 1979; Puhani, 2000). Just inside the favela border, firms select a small number of zip codes for which they accept orders charging lower delivery prices. Farther inside the favela, where there are fewer zip codes available to select from, firms balance a trade-off between accepting orders but increasing the delivery price charged.²⁹

The results support the critical geography argument (Davis, 2006; Melgaço & Prouse, 2017; Santos, 1978, 1987, 2017; Zenou & Boccard, 2000) that spatial inequality perpetuates socioeconomic inequality as manifest in less access to goods and services. The “favela effect” happens as e-commerce firms either refuse orders or charge more for delivery to favela consumers after identifying the consumer location as inside a disadvantaged community.

A heterogeneity test was performed to evaluate whether firms treat favela consumers differently from each other (see Table 1 in the Supporting Information Appendix 2 for more details). The joint-significance tests for heterogeneity in the coefficients for firms confirm that the firms are statistically different in their approaches toward favelas consumers on all dependent variables Y_{jkzt} . Refused orders have a χ^2 test equal to 13.80 (p -value = .032), delivery price has an F -test equal to 2.26 (p -value = .058), and delivery price ratio has an F -test equal to 2.43 (p -value = .044).

6 | MECHANISMS BEHIND FIRM HETEROGENEITY

Qualitative analysis was conducted to explore possible mechanisms behind the quantitative results. The objective is to deepen the understanding of patterns behind firms' heterogeneous behaviors and to lend support to the regression results (Kaplan, 2016; Small, 2011).

²⁷Supporting Information Appendix 8 presents the slope significance analyses for distances outside and inside favelas.

²⁸Based on field observations and interviews with consumers, one possible explanation for the absence of a discontinuity in the delivery price graphs is that firms may deliver the purchases to a reference point on the border of the favela, and the consumer goes there to pick them up. In other words, the discontinuity cut-off point is not strictly adhered to (Angrist & Pischke, 2008).

²⁹Supporting Information Appendix 4 supports this conjecture in an analysis based on a Heckman two-step model estimation (Heckman, 1979; Puhani, 2000). The Heckman model results confirm that firms initially select the zip codes for which they accept orders and subsequently set the delivery price in this sample.

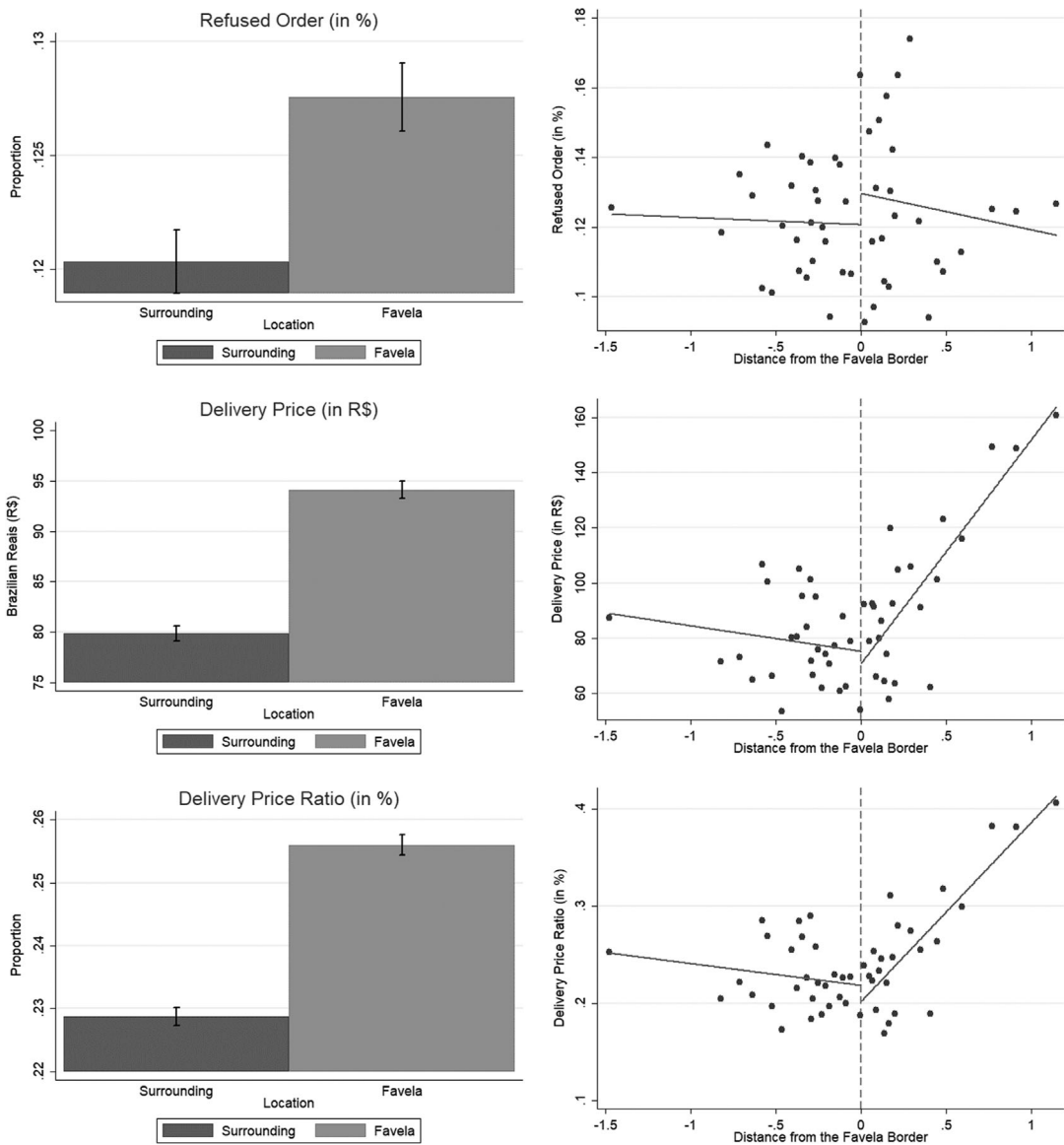


FIGURE 3 “The favela effect.” Parametric and nonparametric estimation of refused order (in %) delivery price (in R\$), and delivery price ratio (in %). The nonparametric estimation was performed considering 50 bins and a discontinuity point at the favela border. All negative distances represent zip codes outside the favelas, and all positive distances zip codes inside the favelas. The graphs were elaborated controlling for firm, week, location, and product category fixed effects. Note that the y-axes of the graphs are on different scales

6.1 | Qualitative data source

Thirteen semi-structured interviews of 30–45 min in length were conducted with various informants, including favela consumers, logistical operators, and managers of physical stores. The selection of the informants was defined by convenience and through a chain sampling

procedure, in which informants sequentially refer to others (Patton, 2005; Strauss & Corbin, 1990). I applied ethnographic techniques while living in *Favela da Rocinha* (Rio de Janeiro) for a week.³⁰ I also visited other favelas in the sample to perform interviews and make field observations (Kaplan, 2008; Pratt, 2003; Pratt & Kim, 2012; Rosen, 1991).

The qualitative analysis reported here focuses on interviews with favela consumers and physical store managers from one e-commerce firm (Firm #03).³¹ This particular e-commerce firm sells products online and operates physical stores inside three favelas. I visited a store in *Favela da Rocinha* (Rio de Janeiro) and another in *Favela de Paraisópolis* (São Paulo).

7 | RESOURCES AND CAPABILITIES—PHYSICAL STORES AS COMPLEMENTARY ASSETS

7.1 | Qualitative evidence about the mechanism

According to interviews and field observations, e-commerce firms with complementary assets—for example, physical stores located within or nearby favelas—may charge favela consumers differently from firms that operate online only. Physical stores inside favelas serve as a delivery hub to facilitate transactions and as a laboratory through which the firm learns how to operate in resource-scarce locations. The informants confirmed that physical stores act as hubs:

Clients can purchase online and come to the store [which is inside the favela] to pick it up.— Store Manager at Favela da Rocinha (Rio de Janeiro)

The complementarity of physical stores is not confined to inside favelas. Physical stores outside favelas also serve this purpose:

My most recent purchases were this watch and some books. And, when I buy books, I buy them in [Firm #09], so I order them for pick up, and I go there to pick them up in their physical store. (...) The store is in Barra da Tijuca [neighborhood 15 km away from the favela].— Consumer living in Favela da Rocinha (Rio de Janeiro)

The quote suggests that the physical store works as a complement to the firm online marketplace.³² The availability of the store as a complementary asset increases consumers' access to goods and may be associated with lower delivery charges. Following Barney's (1991) definition of firm *resources* as: "(...) all assets, capabilities, organizational processes, firm attributes, information, knowledge, etc. controlled by a firm that enables the firm to conceive of and implement strategies that improve its efficiency and effectiveness." (Barney, 1991, p. 101), the qualitative evidence suggests that a firm's *resources and capabilities* may moderate the favela effect (Barney, 1991; Tripsas, 1997).

³⁰As 1 week is a short time to be classified as an ethnography, I state that I have applied "ethnographic techniques" instead.

³¹During the period I was living/visiting two of the favelas in my sample, I had access to store managers and several local consumers. Following research ethics procedures, I have kept the informants' identities confidential.

³²Not all e-commerce firms offer the option of within-store pick up in their marketplace webpages. Among the firms with stores inside favelas, Firm #03 is the only one that offers the option of in-store pick up, which usually posts a delivery price of zero for in-store delivery.

7.2 | A statistical test of the mechanism

In line with the *resources and capabilities* rationale, I test quantitatively whether e-commerce firms with complementary assets—“Physical Stores”—charge less for delivery to customers placing orders to favelas using the following model:

$$Y_{jkzt} = \rho_0 + \rho_1 \text{Favela}_{jkzt} \times \text{Physical Stores}_j + \rho_2 \text{Favela}_{jkzt} + \rho_3 \text{Physical Stores}_j + \rho_4' Z_{jkzt} + \varepsilon_{jkzt} \quad (2)$$

In this specification, two variables are interacted with the favela dummy: “Firm Has Physical Stores” (dummy) and “Firm Has Physical Stores In Favelas” (dummy). The former takes the value of 1 if the e-commerce firm has physical stores and 0 if the firm is only internet-based. The latter takes the value of 1 if the e-commerce firm has physical stores in any favela and 0 otherwise. The former variable measures the effect of complementary assets to support transactions in general, while the latter measures not only transactional support but also the potential for knowledge generation inside the community.³³

Table 4 presents the results of the moderation test. The LOGIT models show that firms with physical stores either outside or inside favelas are less likely to refuse an order from a favela consumer (respectively, -0.180 with $SE = 0.0765$ and $p\text{-value} = .019$; and -0.175 with $SE = 0.0702$ and $p\text{-value} = .013$). Firms with stores in favelas have an equivalent likelihood of refusing an order to a favela consumer as to a non-favela consumer.³⁴ The OLS models indicate that physical stores have no significant effect on the absolute or proportional delivery charge to favela customers. Firms with physical stores in favelas charge R\$ 4.49 (\$ 1.35 US dollars) more to deliver products to favela consumers ($SE = 1.632$, $p\text{-value} = .012$).

These results show that physical stores, either inside or outside favelas, serve as complementary assets that are associated with reductions in order refusals to favela customers. However, these stores may be associated with higher delivery charges. This complex result suggests that physical stores indeed serve as a delivery hub to facilitate transactions. Still, their availability may create an incentive for firms to charge higher for home delivery to compel customers to elect store delivery instead of door delivery. This incentive would arise if pickup from physical stores reduces costs and induces cross-selling to customers, for example. Overall, the analysis supports the *resources and capabilities* rationale.

8 | STAKEHOLDER MANAGEMENT—EMPLOYEES AND CONSUMERS' ASSESSMENT OF THE FIRM

8.1 | Qualitative evidence about the mechanism

Field evidence indicates that firms investing more in their relationships with employees and customers might charge less for delivery in favelas than firms investing less. By investing in a stakeholder-oriented culture—, an organizational set of values and practices oriented toward the well-being of employees and customers—firms may signal their trustworthiness and

³³ Among all the firms in the database, only two firms have physical stores inside favelas (see Supporting Information Appendices 6 and 7). One has one store, and the other has three.

³⁴ The value comes from the ratio between coefficients in Table 4 LOGIT (2), such that: $(-0.175/0.174) = -100\%$.

TABLE 4 *Resources and Capabilities* moderation test—physical stores as complementary assets

	LOGIT (1)	LOGIT (2)	OLS (1)	OLS (2)	OLS (3)	OLS (4)
	1(Refused order)		Delivery price (in R\$)		Delivery price ratio (in %)	
Favela × Firm Has Physical Stores	−0.180 (0.0765)		−0.780 (0.661)		−0.000730 (0.00221)	
Favela × Firm Has Physical Stores in Favelas		−0.175 (0.0702)		4.491 (1.632)		0.00118 (0.00400)
Favela	0.287 (0.0588)	0.174 (0.00992)	19.33 (0.573)	17.67 (0.379)	0.0312 (0.00188)	0.0303 (0.000955)
Firm Has Physical Stores	0.730 (0.0414)		−6.870 (3.285)		−0.00412 (0.00734)	
Firm Has physical Stores in Favelas		−0.774 (0.0959)		−28.27 (4.366)		−0.108 (0.0134)
Week Fixed Effect	Y	Y	Y	Y	Y	Y
Product Category Fixed Effect	Y	Y	Y	Y	Y	Y
Firm Fixed Effect	Y	Y	Y	Y	Y	Y
Location Fixed Effect	Y	Y	Y	Y	Y	Y
Constant	−0.199 (0.129)	0.499 (0.118)	26.09 (8.422)	19.62 (8.978)	0.0652 (0.0206)	0.0612 (0.0186)
Observations	288,679	288,679	354,077	354,077	354,077	354,077
Pseudo and <i>R</i> -squared	0.147	0.147	0.334	0.334	0.385	0.385
Log-Likelihood	−114,008	−114,013				

Note: Standard errors are clustered at 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on 10 different products of four different categories (by size and value), for 11 different e-commerce firms, operating in the main state capitals of Brazil.

commitment to serve disadvantaged communities. Informants described this process by comparing e-commerce firms' practices. According to a store manager, the approach toward favelas reflected the firm's identity of caring about low-income populations as valuable customers:

(...) [Firm #03] is a store from the people. Since Mr. [Firm #03 founder name]'s time, he used to sell blankets to low-income people door-to-door; then, the company scaled up to other social classes. However, this logic of dealing with low-income populations is in the DNA of the company.— Store Manager at *Favela de Paraisópolis* (São Paulo)

The stakeholder orientation of some firms was also identified as essential by local customers. In response to a question about purchasing a cellphone, an informant answered:

I think of [Firm #04] and [Firm #03] because of the convenience. Both deliver 'easily' here. So, they have a logistics that, for example, they know everyone here, they

know the streets and the addresses. (...) They even hire people from here. I had a friend that worked as a delivery guy for [Firm #04].— Consumer living in *Favela de Paraisópolis* (São Paulo)

Several elements of this quote reflect salient ideas in *stakeholder management* theory. First, the consumer mentioned “Firm #03,” which is the employer of the previous informant. This shows consistency in the relationship between the firm with different types of stakeholders (employees and customers). Second, the consumer described her preference for “Firms #03 and #04” because they have superior knowledge of performing deliveries in the favelas, emphasizing the development of specific know-how on transit within this setting. Third, the consumer suggested that superior performance arises from hiring individuals from the favela, reinforcing the firm’s legitimacy and commitment to the community even through employment relationships.

Following Donaldson and Preston’s (1995) definition of firm *stakeholders* as: “(...) persons or groups with legitimate interest in procedural and/or substantive aspect of corporate activity.” (Donaldson & Preston, 1995, p. 67), the qualitative evidence suggests that a firm’s *stakeholder management* may moderate the favela effect (Dorobantu & Odziemkowska, 2017; Lumpkin & Bacq, 2019; Luo & Kaul, 2019).

8.2 | A statistical test of the mechanism

In line with the *stakeholder management* logic, I test whether the “Stakeholder Orientation” of the firm moderates the favela effect following the model:

$$Y_{jkzt} = \rho_0 + \rho_1 \text{Favela}_{jkzt} \times \text{Stakeholder Orientation}_j + \rho_2 \text{Favela}_{jkzt} + \rho_3 \text{Stakeholder Orientation}_j + \rho_4' Z_{jkzt} + \varepsilon_{jkzt} \quad (3)$$

The importance of a stakeholder orientation is measured by two variables: “Employees’ Assessment of the Firm Culture” (scores 1–5) and the firm’s “Responsiveness to Consumer Complaints” (in %). The first variable reflects the degree to which the values and practices of firms are consistent and aligned in the perception of internal stakeholders. The second measure captures the perception of this consistency and alignment by general external stakeholders. The data on employees’ evaluations of firm culture was collected from the hiring website, [Indeed.com](https://indeed.com).³⁵ Recent research using data from hiring websites has proven to be effective in measuring firms’ engagement with internal stakeholders (Brown, 2017; Chang, Kovacs, & Sharkey, 2018; Corritore, Goldberg, & Srivastava, 2020). The data on firm responsiveness to consumer complaints was collected from the consumer rating website, [ReclameAqui.com.br](https://reclameaqui.com.br) (English: “Complaints Here”).³⁶ Prior research has relied on consumer reviews to measure a firm’s commitment to its final clients (Bettinazzi & Feldman, 2020; Hawkins & Hoch, 1992; Kovács & Sharkey, 2014).³⁷

³⁵I collected similar data on a Brazilian hiring website named [Catho.com](https://catho.com). The results are consistent in both cases. I opted for [Indeed.com](https://indeed.com) for the final analyses because it has more reviews (hundreds) and it is a recognized international website.

³⁶Complaints are performed by consumers living inside and outside favelas. The website does not allow access to individual ratings or location. Ratings are aggregated at the firm level over the entire number of complaints performed.

³⁷I tested how aligned both constructs are. The two standardized constructs have a .50 score in Cronbach’s alpha. A Cronbach’s alpha of .50 is considered “poor.” However, there is some discussion as to what would be an adequate cut-off for the Cronbach’s alpha measure (Lance, Butts, & Michels, 2006). Therefore, given the small number of items of the construct, for the purposes of this research, the reliability level is sufficient.

Table 5 presents the results of tests on stakeholder orientation. LOGIT models show that firms with higher ratings from both employees and customers tend to be less likely to refuse an order from favela customers (respectively, -0.323 , with $SE = 0.152$ and $p\text{-value} = .034$; and -0.239 , with $SE = 0.136$ and $p\text{-value} = .079$). OLS models show that firms with one point higher ratings in their culture score by employees charge less for delivery to favela customers both absolutely (R\$ 10.58 less, with $SE = 4.055$ and $p\text{-value} = .017$, which corresponds to \$ 3.18 US dollars less) and proportionately (4.99 p.p. less, with $SE = 0.015$ and $p\text{-value} = .004$). Firms that are rated higher by customers are not associated with lower absolute delivery charges but do charge proportionately less for delivery (1.96 p.p. less, with $SE = 0.006$ and $p\text{-value} = .003$). These results demonstrate that firms with a stakeholder orientation are less likely to refuse an order from a favela customer and charge less to perform the delivery. This charge can be up to 23% lower than the average.³⁸

A supplementary test was conducted in which higher-rated firms were separated from lower-rated ones. This test was conducted on the variable, “Employees’ Assessment of the Firm Culture,” which was assigned a value of 1 for observations associated with the three firms that were rated highest by employees—Firm #03, Firm #09, and Firm #07, respectively—and a value of 0 for all other firms. Both parametric and nonparametric estimations evaluate the likelihood of order refusals, the delivery price, and the delivery price ratio, considering the border of the favela as the discontinuity point.

Figure 4 shows binned scatter plots to lend support to the regression results visually (Starr & Goldfarb, 2020). The non-top-three firms (in green) refuse more orders when transacting with favela customers than with their surrounding counterparts. In contrast, the top three firms (in pink) accept more orders from favela customers than from their surrounding counterparts. All firms have a kink in their pricing strategies toward favelas (Nielsen et al., 2010). Nevertheless, the top-three firms charge significantly less than non-top-three firms. Overall, this constitutes support for *stakeholder management* rationale.

9 | INTEGRATING RESOURCES AND CAPABILITIES WITH STAKEHOLDER MANAGEMENT

Table 6 summarizes the two mechanisms described in the article. Both *resources and capabilities* and *stakeholder management* theories appear relevant for understanding why and how some firms accept more orders and set lower delivery charges in transactions with favela customers.

In line with recent literature (Barney, 2018; McGahan, 2021), Figure 5 presents evidence that an *integration* of both resource-based and stakeholder mechanisms yields greater explanatory power than either alone. This finding is supported by the fact that firms that combine resources (physical stores as supporting hubs) with a stakeholder orientation tend to charge less for delivery to favela consumers ($p\text{-value} = .004$ for cases with stakeholder orientation and stores outside favelas, and $p\text{-value} = .024$ for cases with stakeholder orientation and stores inside favelas).

In particular, consider the case of “Firm #03”,³⁹ which is the firm for which both resource-based and stakeholder measures are greatest. Firm #03 has physical stores both

³⁸The value comes from the ratio between the coefficients in Table 5 OLS (3), such that: $(-0.0499/0.217) = -23\%$.

³⁹I opted to test the integration effect through a case study instead of triple or quadruple interaction terms so as to avoid losing statistical power in the test (type II error).

TABLE 5 *Stakeholder Management* moderation test—employees and consumers' assessment of the firm

	LOGIT (1)	LOGIT (2)	OLS (1)	OLS (2)	OLS (3)	OLS (4)
	1(Refused Order)		Delivery Price (in R\$)		Delivery Price Ratio (in %)	
Favela × Employees' assessment of the firm culture	−0.323 (0.152)		−10.58 (4.055)		−0.0499 (0.0154)	
Favela × Responsiveness to consumer complaints		−0.239 (0.136)		−0.294 (1.547)		−0.0196 (0.00579)
Favela	1.352 (0.569)	23.99 (13.58)	58.07 (15.11)	47.93 (154.0)	0.217 (0.0573)	1.982 (0.576)
Employees' assessment of the firm culture	5.150 (0.136)		−38.72 (16.56)		−0.285 (0.0367)	
Responsiveness to consumer complaints		−2.379 (0.108)		22.02 (8.503)		0.164 (0.0216)
Week fixed effect	Y	Y	Y	Y	Y	Y
Product category fixed effect	Y	Y	Y	Y	Y	Y
Firm fixed effect	Y	Y	Y	Y	Y	Y
Location fixed effect	Y	Y	Y	Y	Y	Y
Constant	−18.55 (0.544)	237.0 (10.83)	162.3 (60.84)	−2,169 (846.9)	1.115 (0.123)	−16.21 (2.166)
Observations	288,679	288,679	354,077	354,077	354,077	354,077
Pseudo and <i>R</i> -squared	0.147	0.147	0.334	0.334	0.386	0.386
Log-Likelihood	−114,018	−114,022				

Note: Standard errors are clustered at 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on 10 different products of four different categories (by size and value), for 11 different e-commerce firms, operating in the main state capitals of Brazil.

outside and inside favelas, and it has one of the highest ratings from both employees and consumers (see Supporting Information Appendices 6 and 7). According to one of the informants, Firm #03's decision to open physical stores inside favelas reflects an intrinsic motivation embedded in the firm's culture to serve disenfranchised consumers. The informant explained that this decision helped the firm reaffirm its role as a change-maker both with employees and favela consumers. This strategy increased the firms' legitimacy and trustworthiness with both groups. The dual objective of the firm to both operate profitably and promote the availability of products for favela dwellers was acknowledged by the manager. In her words:

In the past, these places [favelas] did not have this type of store. So the company innovated; did something different. They thought: let's open inside the favela! Then now, we have, for example, here at Paraisópolis [São Paulo], another one in

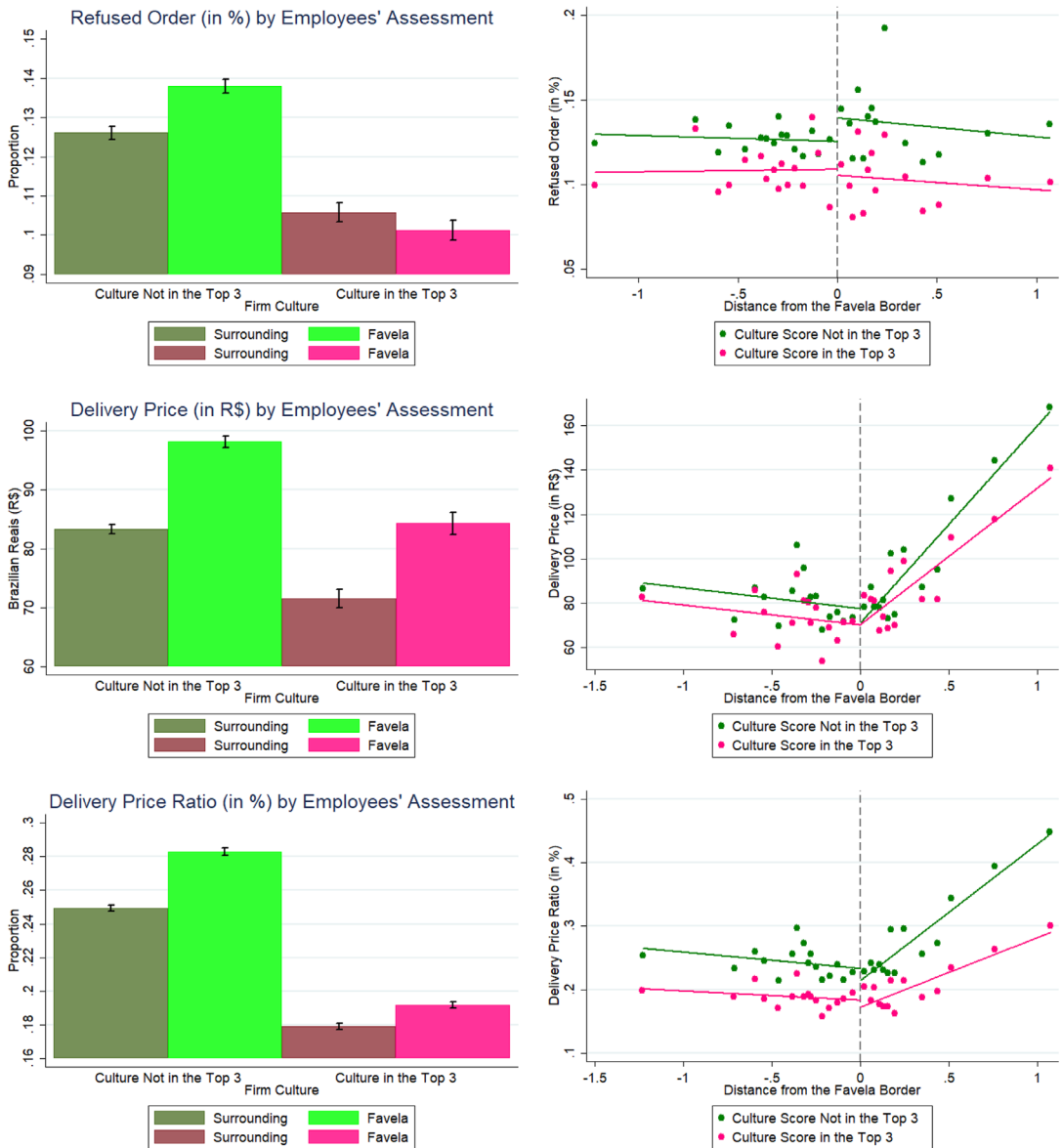


FIGURE 4 *Stakeholder Management* moderation effect, based on the employees' assessment of the firm culture (top three scores). Parametric and nonparametric estimation of refused order (in %) delivery price (in R\$), and delivery price ratio (in %). The nonparametric estimation was performed considering 30 bins and a discontinuity point at the favela border. All negative distances represent zip codes outside the favelas, and all positive distances zip codes inside the favelas. The graphs were elaborated controlling for firm, week, location, and product category fixed effects. Note that the y-axes of the graphs are on different scales

Heliópolis [São Paulo], and also in Rocinha [Rio de Janeiro]. So it was a good thing for the clients and us. I think it was necessary, a necessary change.

—Store Manager at *Favela de Paraisópolis* (São Paulo)

TABLE 6 Summary of the rationale for firms' heterogeneity

	Resources and capabilities	Stakeholder management
Rationale for the firm behavior	Complementary assets serve as delivery hubs and laboratories, leading the firm to reach more and charge less of disenfranchised consumers.	A stakeholder-oriented culture increases the firm trustworthiness and legitimacy in disadvantaged communities, leading the firm to reach more and charge less of disenfranchised consumers.
Source of heterogeneity	Physical stores both outside and inside favelas.	Employees and consumers' assessment of the firm.
Qualitative evidence from field visits and interviews	"I order them (books) for pick up, and I go there to pick them up in their physical store (located 15 km away from the favela)." "Clients can purchase online and come to the store (inside the favela) to pick it up."	"(...) [firm #03] is a store from the people. (...) this logic of dealing with low-income populations is in the DNA of the company." "They [firm #03 and #04] know everyone here, they know the streets and the addresses. (...) they even hire people from here. I had a friend that worked as a delivery guy for [firm #04]."
Measures used in the quantitative analysis	• The firm has physical stores (dummy); • The firm has physical Stores in Favelas (dummy).	• Employees' assessment of the firm culture (scores 1–5); • Firm responsiveness to consumer complaints (in %).
Results from the quantitative analysis	• Support heterogeneity of refused orders; • Partially reject heterogeneity of delivery price; • Null effect on heterogeneity of delivery price ratio.	• Support heterogeneity of refused orders; • Partially support heterogeneity of delivery price; • Support heterogeneity of the delivery price ratio.
Interpretation of the results	The presence of complementary assets, on average, increases the likelihood of accepting an order from a favela zip code up to 100%. However, there are null or partially increasing effects in terms of delivery price and delivery price ratio.	A stakeholder-oriented approach not only increases the likelihood of accepting an order from a favela zip code but also reduces the delivery price and delivery price ratio charged to favela consumers up to 23%, on average.

A final test supports the integration argument. This test examines whether and how much the “Firm #03” (dummy) moderates the favela effect (see Tables 7 and 8 in Supporting Information Appendix 3 for more details).

The results show that the likelihood that a favela customer's order is refused by Firm #03 is comparable to the likelihood of refusal to an order from a non-favela customer (p -value = .013).⁴⁰ In other words, the firm makes goods available within favelas and outside favelas commensurately. Furthermore, Firm #03's delivery charges to favela customers are 45% lower than

⁴⁰The value comes from the ration between coefficients in Table 7, Supporting Information Appendix 3 LOGIT (1), such that: $(-0.175/0.174) = -100\%$.



FIGURE 5 Integration of resources and capabilities and stakeholder management. Baseline on “no stores nor stakeholder orientation” case. The estimation was performed on the interaction effect between the firm’s case type and favela dummy on the delivery price ratio variable. The graph shows the point estimate and 95% confidence interval controlling for favela, firm case type, week, location and product category fixed effects. Supporting Information Appendix 6 presents the classification of firms by case types

those of competitors (p -value = .015).⁴¹ In sum, Firm #03 uses its resources and manages its stakeholders to develop cheaper alternatives for commercialization and distribution within favelas.

Finally, interviewed informants explained that this approach simultaneously supported local communities and created economic benefits for the firm by enlarging its market share in the favelas. In the words of the informants:

In comparison with Gávea [wealthy neighborhood located in the surroundings of Rocinha favela], where there is a store like this one, our store sells at least R\$ 500 to R\$ 600 thousand⁴² more than theirs.— Store Manager at Favela da Rocinha (Rio de Janeiro)

Today, the profit here [in the store inside the Paraisópolis favela] is bigger than outside. For example, some stores sell R\$ 1.3 million, and others R\$ 1.8 million⁴³; our store is more in this second group.— Store Manager at Favela de Paraisópolis (São Paulo)

Overall, these results reinforce the simultaneous relevance of resource-based and stakeholder explanations, particularly for firms operating in disadvantaged communities.

⁴¹The value comes from the ration between coefficients in Table 7, Supporting Information Appendix 3 OLS (2), such that: $(-0.0143/0.0321) = -44.5\%$.

⁴²Corresponds to \$ 150,000–\$ 180,000 US dollars.

⁴³Corresponds to \$ 390,000–\$ 540,000 US dollars.

10 | ROBUSTNESS CHECKS

Other than the RDD suitability tests previously mentioned, I performed the following additional analyses to assess the robustness of the results: (i) firm and product category heterogeneity tests (see Supporting Information Appendix 2), (ii) alternative dependent variables tests measuring products classified as “out of stock” (dummy), and the “delivery time” (in days) of accepted transactions (see Supporting Information Appendix 3), (iii) a Heckman test (Heckman, 1979; Puhani, 2000) considering possible selection bias that firms insert in the sample by picking the zip codes they are willing to accept orders from, before setting the delivery price (see Supporting Information Appendix 4), (iv) a formal prediction model of delivery pricing for favela consumers (see Supporting Information Appendix 5), (v) assessments of correlations between firms’ characteristics and estimated revenues (see Supporting Information Appendix 6), (vi) a slope significance analysis to verify the regression discontinuity kink results (Nielsen et al., 2010) (see Supporting Information Appendix 8), (vii) a bandwidth analysis varying the range of zip-codes from 750 to 50 m (equivalent from 0.47 to 0.03 miles) distance to the threshold (see Supporting Information Appendix 9), and (viii) additional OLS tests for the binary outcomes (see Supporting Information Appendix 12). The results of these additional tests are consistent with the main findings described in the article.

11 | DISCUSSION

This phenomenological study investigates firm strategies in disadvantaged urban communities. In line with the literature on critical geography and urban economics (Chown & Liu, 2015; Chyn, 2018; Davis, 2006; Melgaço & Prouse, 2017; Santos, 1978, 1987, 2017; Zenou & Boccoard, 2000), the study considers an often overlooked issue in strategic management—the geographical space—in the assessment of discrimination of disenfranchised individuals by firms. Following this literature, the analysis considers how firms treat customers differently based on their zip codes even when no evidence arises of differences in delivery costs to closely located neighborhoods. The study evaluates how spatial inequality manifests in differences in the treatment of favela and non-favela customers, which perpetuates socioeconomic disparity in access to products and services. The analysis reports on a large and unique dataset of 406,900 transaction quotes between e-commerce firms and customers inside and outside 21 Brazilian favelas. Quantitative results show that firms are more likely to refuse orders and charge more for delivery to customers living in favelas than immediately outside of favelas. This finding reflects the “favela effect” through which spatial inequality translates into socioeconomic disparity as firms charge higher prices to low-income consumers living in communities defined by scarcity of resources. The study finds heterogeneity in pricing among firms. Some firms charge favela consumers less than others. This heterogeneous behavior reflects that some firms have developed strategies that enable better service to disadvantaged communities than other firms. Firms that rely on physical stores and that cultivate relationships with both employees and consumers charge favela customers more equitably than competing firms.

These findings contribute to strategic management in several ways. First, the study develops the literature on *resources and capabilities* (Barney, 1991, 2001; Kogut & Zander, 1992; Nickerson & Zenger, 2004; Peteraf, 1993; Silverman, 2002; Tripsas, 1997; Wernerfelt, 1984). The article reveals the importance of complementary assets supporting a firm’s operations in disadvantaged communities. These assets serve as hubs facilitating transactions and as vehicles

through which learning and tacit knowledge are obtained. Knowledge about effective operations in resource-scarce settings may be transferable and accumulated intertemporally, reinforcing the firm's competitive advantage. The practitioner literature has shown some evidence of this process. For example, e-commerce firms such as Alibaba in China and ITC e-Choupal in India have invested in physical shops and kiosks as complementary assets to the online market. These assets supported the accumulation of knowledge about serving disenfranchised consumers to achieve superior economic gains (Leong, Pan, Newell, & Cui, 2016; Narsalay, Pongeluppe, & Light, 2015; Prahalad, 2006). The findings in this article reinforce the argument that firms can overcome spatial inequalities and better serve disenfranchised consumers if they redeploy their resources and capabilities.

Second, the study contributes to the *stakeholder management* literature (Amis, Barney, Mahoney, & Wang, 2020; Donaldson & Preston, 1995; Dorobantu & Odziemkowska, 2017; Henisz et al., 2014; Jones, 1995; Lumpkin & Bacq, 2019; Luo & Kaul, 2019) by showing that a stakeholder orientation leads firms to treat consumers living in disadvantaged communities equitably. It also shows that, regardless of intention,⁴⁴ firms with a culture of care toward employees and consumers enhance their trustworthiness (McEvily et al., 2003) and social legitimacy (Ingram & Silverman, 2016; Webb, Tihanyi, Ireland, & Sirmon, 2009) in disadvantaged areas. Several examples illustrate the potential generalizability of these results. Telecommunication firms such as Vodafone in India, Zain in Kenya, and Celtel in Nigeria have promoted employee commitment (i.e., “buy-in”) and pursued a trustworthy reputation within disadvantaged communities to achieve better operational and financial results in those settings (Anderson, Markides, & Kupp, 2010). Similarly, Casas Bahia, a retail firm in Brazil, has developed a culture of serving poor customers in ways that reflect their resource constraints (Prahalad, 2006, pp. 119–120). The findings in this article emphasize the claim that firms with a stakeholder orientation may see disenfranchised individuals as valuable customers and employees. In doing so, these firms engage in more genuine relationships with these individuals, contributing to the enfranchisement of disadvantaged communities.

Third, the study adds to a recent development in the strategic management literature (Barney, 2018; McGahan, 2021), showing that resource-based and stakeholder management explanations are highly consistent. The article demonstrates that integrating resources and stakeholder orientation is critical when firms operate in resource-scarce settings. While resource redeployment increases a firm's efficiency in resource-scarce locations, a stakeholder orientation increases a firm's legitimacy and trustworthiness within the community. Combined, they contribute simultaneously to the inclusion and empowerment of disadvantaged communities and assure a superior competitive position for the firm.

This article also suggests several meaningful opportunities for future research in the field of strategic management. Theoretically, the article highlights the importance of further studies to understand the symbiotic interaction between organizations' resources and stakeholders, and the consequences of this iterative relationship. Phenomenologically, the article emphasizes the importance of broadening the range of dependent and independent variables of interest. Strategic management researchers have the opportunity to extend their conceptual realm above and beyond the drivers of firm economic performance. Outcomes related to organizations' social

⁴⁴The reasons for a stakeholder orientation include moral imperatives for inclusion (Kaplan, 2019; Lazzarini, 2020; Rangan, 2018) and the expectations of competitive economic gains in the long-run (Cheng, Ioannou, & Serafeim, 2014; Dorobantu & Odziemkowska, 2017; Eccles et al., 2014; Flammer & Kacperczyk, 2016; Flammer, 2015; Luo & Kaul, 2019; Nardi, 2021).

and environmental performances can and should be further studied and theorized. Methodologically, the field may benefit from further use of mixed methods. The triangulation of information and the holistic understanding of phenomena through the interplay of quantitative and qualitative analyses can improve the epistemological interpretation of complex problems. This is particularly true for historically underdeveloped research areas in which data cannot be readily found—for example, the grand societal challenges, the SDGs, and the Global South.

This study also carries the potential to inform practitioners and policy-makers. Practitioners may learn that operations in disadvantaged locations demand specific investments that support transactions and contribute to the realization of the underdeveloped potential of those communities. The results highlight the importance of building an internal culture of care toward disenfranchised stakeholders that supports the development of trust. In doing so, firms gain social legitimacy and genuinely engage with the community and its constraints. The results suggest that operating and succeeding in commercial endeavors within disadvantaged communities require resource deployment and the cultivation of a stakeholder culture that enables the change-making of structural inequalities that deepen impoverishment.

Policy-makers may rely on the results of this study to verify the importance of regulatory constraints on firm behavior. This is particularly important for shaping the activity of online firms, such as e-commerce firms, given the considerable asymmetry in information, bargaining power, and voice between these firms and their consumers. The study also supports policy-makers in recognizing how spatial inequalities, such as those between favelas and surroundings, become complicit through firm behavior in deepening inequalities between closely located communities. Policy-makers may seek to restrain spatial inequalities and the ways they manifest into other forms of inequalities related to economic access to goods, services, and opportunities.

11.1 | Limitations

There are three main limitations of this article. First, informal negotiations with local delivery carriers may occur within favelas. This alternative arises when the e-commerce firm delivers at the favela border (see Supporting Information Appendix 5) or at a drop-off place inside the favela.⁴⁵ As the quantitative models cannot capture the “total price” consumers pay to informal delivery carriers, there is a possibility that the estimators of delivery charges are biased downward. This consideration is of theoretical importance. Favelas are rich ecosystems in which formal and informal agents interact, collaborate, and compete (Lenz, Sutter, Goldszmidt, & Zucco, 2021). By focusing on the interaction between e-commerce firms and local consumers, the article might be ignoring important economic and societal outcomes which are theoretically interesting but have not been thoroughly observed and investigated in this study.

⁴⁵According to the informants, in these cases, the consumer has to bear the cost of bringing their purchase from the drop-off point to their residence. Thus, the customer has to negotiate with local carriers, who perform the last mile of delivery. Based on the evidence provided by the informants and the field observation, these local carriers are usually unemployed individuals who live in the favela. These individuals performing last-mile delivery charge the final consumer a price for the service. The price of this last-mile of service varies from R\$ 25 Brazilian reais to R\$ 100 Brazilian reais (from \$ 7 to \$ 25 U.S. dollars) depending on the size of the product, and the distance between the drop-off place and the consumer home. In some cases, this amount of money may represent up to 37.8% of the consumer's monthly income (see Supporting Information Appendix 1).

Second, potential biases in reporting crimes inside favelas cannot be ruled out. The lack of insurance against theft in the favelas might affect the likelihood of reporting some types of crimes. Also, generalized social myths and beliefs of higher criminality in the favelas may alone be sufficient to drive different firm behavior inside and outside of them. Unfortunately, the article cannot access the belief systems embedded in e-commerce firms' algorithms.

Third, the peculiarities of the Brazilian context might hamper the generalizability of the results to other institutional settings. For example, in Brazil, favela populations are subject to social discrimination and criminalization even by state forces (e.g., some politicians and police forces). The spatial proximity between favelas and wealthier communities is particularly remarkable in Brazil. Nevertheless, the findings may be relevant in other institutional settings where social stigmatization of the poor, and income inequality in urban centers are prevalent.

12 | CONCLUSION

This article seeks to contribute to a new paradigm in management that aims to understand profoundly important societal problems. The study adds to the research agenda on strategic management by bringing insights from critical geography and urban economics into the field of strategy. Specifically, this study identifies "the favela effect" through which spatial inequalities reinforce and perpetuate socioeconomic disparities. The favela effect arises when e-commerce firms refuse orders and charge higher prices to customers in disadvantaged urban communities after learning of the spatial location of a customer.

The article points to opportunities for enriching core constructs and bridging gaps among theories within the strategic management field. *Integrating* resource-based and stakeholder management theories can lead to significant insights into how firms remediate socioeconomic inequalities and improve performance (McGahan, 2021). Finally, the article reinforces the claim that firms face important opportunities to contribute to the achievement of the United Nation's Sustainable Development Goals of ending poverty (SDG #01), reducing inequalities (SDG #10), and fostering sustainable cities and communities (SDG #11).

ACKNOWLEDGEMENT

I am deeply grateful to Caroline Flammer (the Editor), two anonymous reviewers, Ruth Aguilera, Celso Athayde, Andy Back, Daphné Baldassari, Sandra Barbosa, Sandro Cabral, Kenneth Corts, Laura Derksen, Alban Drouet, André Duarte, Alberto Galasso, Carlos Inoue, Nobuiuki Ito, Sarah Kaplan, Nicola Lacetera, Sérgio Lazzarini, Kátia Maeda, Pedro Makhoul, Anita McGahan, Aldo Musacchio, Bill McEvily, Francisco Muzetti, Patrick Rooney, Yasser Saleh, Brian Silverman, Thomaz Teodorovicz, as well as participants in sessions at the Academy of Management, the Alliance for Research on Corporate Sustainability (ARCS), the ARCS Ph.D. Sustainability Academy, and the Insper Metricis for their valuable comments and suggestions. This research is exempt from the research ethics review since it involves observation of people in public places, and the dissemination of the research results does not allow the identification of specific individuals. All remaining errors and omissions are my own responsibility.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request. The data are not publicly available due to privacy or ethical restrictions.

ORCID

Leandro S. Pongeluppe  <https://orcid.org/0000-0001-6195-4455>

REFERENCES

- Acquisti, A., Brandimarte, L., & Loewenstein, G. (2015). Privacy and human behaviour in the age of information. *Science*, 347(6221), 509–515.
- Allcott, H., Diamond, R., Dubé, J.-P., Handbury, J., Rahkovsky, I., & Schnell, M. (2019). Food deserts and the causes of nutritional inequality. *Quarterly Journal of Economics*, 134(4), 1793–1844.
- Amis, J., Barney, J., Mahoney, J. T., & Wang, H. (2020). Why we need a theory of stakeholder governance—And why this is a hard problem. *Academy of Management Review*, 45(3), 499–503.
- Anderson, J., Markides, C., & Kupp, M. (2010). The last frontier: Market creation in conflict zones, deep rural areas, and urban slums. *California Management Review*, 52(4), 6–28.
- Angrist, J. D., & Pischke, J.-S. (2008). *Mostly harmless econometrics: An empiricist's companion*. Princeton, NJ: Princeton University Press.
- Bansal, P., & DesJardine, M. (2014). Business sustainability: It is about time. *Strategic Organization*, 12(1), 70–78.
- Barney, J. B. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99–120.
- Barney, J. B. (2001). Is the resource-based “view” a useful perspective for strategic management research? Yes. *Academy of Management Review*, 26(1), 41–56.
- Barney, J. B. (2018). Why resource-based theory's model of profit appropriation must incorporate a stakeholder perspective. *Strategic Management Journal*, 39(13), 3305–3325.
- Battilana, J., & Dorado, S. (2010). Building sustainable hybrid organizations: The case of commercial microfinance organizations. *Academy of Management Journal*, 53(6), 1419–1440.
- Beaulac, J., Kristjansson, E., & Cummins, S. (2009). A systematic review of food deserts, 1966–2007. *Preventing Chronic Disease*, 6(3), 1–10.
- Bellego, C., & Drouard, J. (2019). Does it pay to fight crime? Evidence from the pacification of slums in Rio de Janeiro. *SSRN Electronic Journal*.
- Bertels, S., Howard-Grenville, J., & Pek, S. (2016). Cultural molding, shielding, and shoring at oilco: The role of culture in the integration of routines. *Organization Science*, 27(3), 573–593.
- Bettinazzi, E. L. M., & Feldman, E. (2020). Stakeholder orientation and divestiture activity. *Academy of Management Journal*, 64(4), 1078–1096.
- Bitler, M., & Haider, S. J. (2011). An economic view of food deserts in the United States. *Journal of Policy Analysis and Management*, 30(1), 153–176.
- Brown, D. A. (2017). Toward the hybridization of value: Firm value orientations and collective stakeholder outcomes. *Academy of Management*, 2017, 14497.
- Cameron, A. C., & Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317–372.
- Cameron, A. C., & Trivedi, P. K. (2010). *Microeconometrics using stata* (Revised ed.). College Station, TX: Stata Press.
- Campa, P., & Serafinelli, M. (2019). Politico-economic regimes and attitudes: Female workers under state socialism. *Review of Economics and Statistics*, 101(May), 233–248.
- Chang, M., Kovacs, B., & Sharkey, A. (2018). Valence in organizational culture: Cultural alignment, employee satisfaction, and market performance. *Academy of Management*, 2018, 10844.
- Cheng, B., Ioannou, I., & Serafeim, G. (2014). Corporate social responsibility and access to finance. *Strategic Management Journal*, 35(1), 1–23.
- Chown, J. D., & Liu, C. C. (2015). Geography and power in an organizational forum: Evidence from the US senate chamber. *Strategic Management Journal*, 36, 177–196.
- Chyn, E. (2018). Moved to opportunity: The long-run effect of public housing demolition on labor market outcomes of children. *American Economic Review*, 108(10), 3028–3056.
- Corritore, M., Goldberg, A., & Srivastava, S. B. (2020). Duality in diversity: How intrapersonal and interpersonal cultural heterogeneity relate to firm performance. *Administrative Science Quarterly*, 65(2), 359–394.
- da Cunha, E. (1905). *Os Sertões: Campanha de Canudos* (3a ed.). Rio de Janeiro: Laemmert C.
- Dasgupta, P. (1997). Nutritional status, the capacity for work, and poverty traps. *Journal of Econometrics*, 77, 5–37.

- Davis, M. (2006). *Planet of slums*. London and New York: Verso.
- Duarte, A. L. C. M., Macau, F., Flores e Silva, C., & Sanches, L. M. (2019). Last mile delivery to the bottom of the pyramid in Brazilian slums. *International Journal of Physical Distribution & Logistics Management*, 49(5), 473–491.
- Donaldson, T., & Preston, L. E. (1995). The stakeholder theory of the corporation: Concepts, evidence, and implications. *Academy of Management Review*, 20(1), 65–91.
- Dorobantu, S., & Odziemkowska, K. (2017). Valuing stakeholder governance: Property rights, community mobilization, and firm value. *Strategic Management Journal*, 38, 2682–2703.
- Dutt, N., Hawn, O., Vidal, E., Chatterji, A., McGahan, A., & Mitchell, W. (2016). How open system intermediaries address institutional failures: The case of business incubators in emerging-market countries. *Academy of Management Journal*, 59(3), 818–840.
- Eccles, R. G., Ioannou, I., & Serafeim, G. (2014). The impact of corporate sustainability on organizational processes and performance. *Management Science*, 60(11), 2835–2857.
- Flammer, C. (2015). Does corporate social responsibility lead to superior financial performance? A regression discontinuity approach. *Management Science*, 61(11), 2549–2568.
- Flammer, C., & Bansal, P. (2017). Does a long-term orientation create value? Evidence from a regression discontinuity. *Strategic Management Journal*, 38, 1827–1847.
- Flammer, C., & Ioannou, I. (2021). Strategic management during the financial crisis: How firms adjust their strategic investments in response to credit market disruptions. *Strategic Management Journal*, 42(7), 1275–1298.
- Flammer, C., & Kacperczyk, A. (2016). The impact of stakeholder orientation on innovation: Evidence from a natural experiment. *Management Science*, 62(7), 1982–2001.
- Flammer, C., & Luo, J. (2017). Corporate social responsibility as an employee governance tool: Evidence from a quasi-experiment. *Strategic Management Journal*, 38, 163–183.
- Freyre, G. (1934). *Casa-Grande e Senzala*. Global Editora e Distribuidora Ltda (48th ed.). São Paulo, Brazil: Global Editora e Distribuidora Ltda.
- Freyre, G. (1936). *Sobrados e Mucambos*. São Paulo, Brazil: Brasileira.
- Galhotra, S., Brun, Y., & Meliou, A. (2017). Fairness testing: Testing software for discrimination. *Proceedings of the 2017 11th Joint Meeting on Foundations of Software Engineering, ACM*, pp. 498–510.
- George, G., Howard-Grenville, J., Joshi, A., & Tihanyi, L. (2016). Understanding and tackling societal grand challenges through management research. *Academy of Management Journal*, 59(6), 1880–1895.
- George, G., McGahan, A. M., & Prabhu, J. (2012). Innovation for inclusive growth: Towards a theoretical framework and a research agenda. *Journal of Management Studies*, 49(4), 661–683.
- Gordon, C., Purciel-Hill, M., Ghai, N. R., Kaufman, L., Graham, R., & Van Wye, G. (2011). Measuring food deserts in New York City's low-income neighborhoods. *Health and Place*, 17(2), 696–700.
- Hahn, J., Todd, P., & Van der Klaauw, W. (2001). Identification and estimation of treatment effects with a regression-discontinuity design. *Econometrica*, 69(1), 201–209.
- Hambrick, D. C. (2007). The field of management's devotion to theory: Too much of a good thing? *Academy of Management Journal*, 50(6), 1346–1352.
- Hawkins, S. A., & Hoch, S. J. (1992). Low-involvement learning: Memory without evaluation. *Journal of Consumer Research*, 19, 212–216.
- Heckman, J. (1979). Sample specification bias as a selection error. *Econometrica*, 47(1), 153–162.
- Helfat, C. E. (2007). Stylized facts, empirical research and theory development in management. *Strategic Organization*, 5(2), 185–192.
- Henisz, W. J., Dorobantu, S., & Nartey, L. J. (2014). Spinning gold: The financial returns to stakeholder engagement. *Strategic Management Journal*, 35, 1727–1748.
- Imbens, G. W., & Lemieux, T. (2008). Regression discontinuity designs: A guide to practice. *Journal of Econometrics*, 142(2), 615–635.
- Ingold, D., & Soper, S. (2016). *Amazon doesn't consider the race of its customers. Should it?*. Retrieved from <https://www.bloomberg.com/graphics/2016-amazon-same-day/>
- Ingram, P., & Silverman, B. (2016). The cultural contingency of structure: Evidence from entry to the slave trade in and around the abolition movement. *American Journal of Sociology*, 122(3), 755–797.
- Jones, T. M. (1995). Instrumental stakeholder theory: A synthesis of ethics and economics. *Academy of Management Review*, 20(2), 404–437.

- Jovchelovitch, S., & Priego-Hernandez, J. (2013). *Underground sociabilities: Identity, culture and resistance in Rio de Janeiro's favelas*. Brasília: UNESCO.
- Kaplan, S. (2008). Framing contests: Strategy making under uncertainty. *Organization Science*, 19(5), 729–752.
- Kaplan, S. (2016). Mixing quantitative and qualitative research. In K. D. Elsbach & R. M. Kramer (Eds.), *Handbook of qualitative organizational research: Innovative pathways and methods* (pp. 423–433). New York, NY: Routledge.
- Kaplan, S. (2019). Guidepost trilogy—claims on the corporation: Directions for stakeholder research in the field of management—beyond the business case for social responsibility. *Academy of Management Discoveries*, 6(1), 1–4.
- Kogut, B., & Zander, U. (1992). Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3(3), 383–397. <https://doi.org/10.1287/orsc.3.3.383>
- Kovács, B., & Sharkey, A. J. (2014). The paradox of publicity: How awards can negatively affect the evaluation of quality. *Administrative Science Quarterly*, 59(1), 1–33.
- Lance, C. E., Butts, M. M., & Michels, L. C. (2006). The sources of four commonly reported cutoff criteria; what did they really say? *Organizational Research Methods*, 9(2), 202–220.
- Larkins, E. R. (2015). *The spectacular favela: Violence in modern Brazil*. Oakland, CA: University of California Press.
- Lazzarini, S. G. (2020). The nature of the social firm: Alternative organizational forms for social value creation and appropriation. *Academy of Management Review*, 45(3), 620–645.
- Lee, D. S., & Lemieux, T. (2010). Regression discontinuity designs in economics. *Journal of Economic Literature*, 48(2), 281–355.
- Lenz, A. K., Sutter, C., Goldszmidt, R., & Zucco, C. (2021). Venture distress and problemistic search among entrepreneurs in Brazilian favelas. *Journal of Business Venturing*, 36(6), 106162.
- Leong, C., Pan, S. L., Newell, S., & Cui, L. (2016). The emergence of self-organizing E-commerce ecosystems in remote villages of China: A tale of digital empowerment for rural development. *Management Information Systems Quarterly*, 40(2), 475–484.
- Lucci, P., Bhatkal, T., & Khan, A. (2018). Are we underestimating urban poverty? *World Development*, 103, 297–310.
- Lumpkin, T., & Bacq, S. (2019). Civic wealth creation: A new view of stakeholder engagement and societal impact. *Academy of Management Perspectives*, 33(4), 383–404.
- Luo, J., & Kaul, A. (2019). Private action in public interest. The comparative governance of social issues. *Strategic Management Journal*, 40(4), 476–502.
- Marx, B., Stoker, T., & Suri, T. (2013). The economics of slums in the developing world. *Journal of Economic Perspectives*, 27(4), 187–210.
- McEvily, B., Perrone, V., & Zaheer, A. (2003). Trust as an organizing principle. *Organization Science*, 14(1), 91–103.
- McGahan, A. M. (2018). 2017 presidential address freedom in scholarship: Lessons from Atlanta. *Academy of Management Review*, 43(2), 173–178.
- McGahan, A. M. (2021). Integrating insights from the resource-based view of the firm into the new stakeholder literature. *Journal of Management*, 40(4), 476–502.
- Meirelles, R., & Athayde, C. (2014). *Um país chamado favela*. São Paulo, Brazil: Editora Gente Liv and Edit Ltd.
- Melgaço, L., & Prouse, C. (2017). *Milton Santos: A pioneer in critical geography from the global south*. London: Springer.
- Moore, C. T. (2009). Spatial regression discontinuity: Estimating effects of geographically implemented programs and policies. In *annual conference of the American Evaluation Association*, Orlando, FL.
- Nardi, L. (2021). The corporate social responsibility price premium as an enabler of substantive CSR. *Academy of Management Review*, 47(2), 282–308.
- Narsalay, R., Pongeluppe, L., & Light, D. (2015). The hidden pitfalls of inclusive innovation. *Stanford Social Innovation Review*, Winter(Winter), pp. 48–53.
- Nickerson, J. A., & Zenger, T. R. (2004). A knowledge based theory of the firm: A problem solving approach. *Organization Science*, 15(6), 617–632.
- Nielsen, H. S., Sørensen, T., & Taber, C. (2010). Estimating the effect of student aid on college enrollment: Evidence from a government grant policy reform. *American Economic Journal: Economic Policy*, 2(2), 185–215.

- Patton, M. Q. (2005). Qualitative research. In *Encyclopedia of statistics in behavioral science* (pp. 1633–1636). Hoboken, NJ: John Wiley & Sons, Ltd.
- Perlman, J. (2010). *Favela: Four decades of living on the edge in Rio de Janeiro*. New York: Oxford University Press.
- Peteraf, M. A. (1993). The cornerstones of competitive advantage: A resource-based view. *Strategic Management Journal*, 14(3), 179–191.
- Pless, N. M., & Maak, T. (2004). Building an inclusive diversity culture: Principles. *Journal of Business Ethics*, 54(2), 129–147.
- Prahalad, C. K. (2006). *Fortune at the bottom of the pyramid: Eradicating poverty through profits*. Philadelphia, PA: Wharton School Publishing.
- Pratt, M. G. (2003). Access as relating: On the relationship aspect of different types of access. In M. S. Feldman, J. Bell, & M. T. Berger (Eds.), *Gaining access: A practical and theoretical guide for qualitative researchers*. Walnut Creek, CA: Rowman Altamira Press.
- Pratt, M. G., & Kim, N. (2012). Designing for drift: Planning ethnographic qualitative research on groups. In A. Hillingshead & M. Poole (Eds.), *Research methods for studying groups and teams*. New York and London: Routledge.
- Puhani, P. A. (2000). The Heckman correction for sample selection and its critique. *Journal of Economic Surveys*, 14(1), 53–68.
- Rangan, S. (2018). Capitalism beyond mutuality? In S. Rangan (Ed.), *Capitalism beyond mutuality?: Perspectives integrating philosophy and social science* (p. 1). Oxford, UK: Oxford University Press.
- Rosen, M. (1991). Coming to terms with the field: Understanding and doing organizational ethnography. *Journal of Management Studies*, 28(1), 1–24.
- Santos, M. (1978). *Pobreza Urbana*. São Paulo: Editora USP.
- Santos, M. (1987). *O Espaço do Cidadão*. São Paulo: Editora USP.
- Santos, M. (2017). *Toward an other globalization: From the single thought to universal conscience*. London: Springer.
- SBVC. (2016). Ranking SBVC 50 Maiored Empresas E-commerce Brasileiro 2016. São Paulo, Brazil.
- Shiller, B. R. (2014). *First-degree price discrimination using big data*. Retrieved from http://www8.gsb.columbia.edu/faculty-research/sites/faculty-research/files/finance/Industrial/BenShiller-Nov2014_0.pdf
- Silverman, B. S. (2002). Organizational Economics. In J. A. C. Baum (Ed.), *Blackwell companion to organizations*. London: Blackwell Press.
- Small, M. L. (2011). How to conduct a mixed methods study: Recent trends in a rapidly growing literature. *Annual Review of Sociology*, 37(1), 57–86.
- Starr, E., & Goldfarb, B. (2020). Binned scatterplots: A simple tool to make research easier and better. *Strategic Management Journal*, 41, 1–14.
- Strauss, A. L., & Corbin, J. M. (1990). *Basics of qualitative research: Basics of qualitative research: Techniques and procedures for developing grounded theory* (15th ed.). Newbury Park, CA: Sage.
- Tihanyi, L. (2020). From “Thats Interesting” to “Thats Important”. *Academy of Management Journal*, 63(2), 329–331.
- Tripsas, M. (1997). Unraveling the process of creative destruction: Complementary assets and incumbent survival in the typesetter industry. *Strategic Management Journal*, 18, 119–142.
- UNDP. (2010). *Human development report 2010*. New York, NY: Palgrave Macmillan.
- UN-Habitat (2016). *Slum almanac 2015–2016: Tracking improvement in the lives of slum dwellers*. Independent Review. Nairobi, Kenya.
- Villas Bôas, B. (2020). *Favelas têm poder de compra de R\$ 119,8 bi. Valor Econômico*. Retrieved from <https://valor.globo.com/brasil/noticia/2020/01/24/favelas-tem-poder-de-compra-de-r-1198-bi.ghtml>
- Webb, J. W., Tihanyi, L., Ireland, R. D., & Sirmon, D. G. (2009). You say illegal, I say legitimate: Entrepreneurship in the informal economy. *Academy of Management Review*, 34(3), 492–510.
- Wernerfelt, B. (1984). A resource-based view of the firm. *Strategic Management Journal*, 5(2), 171–180.
- Willis, G. D., & Prado, M. M. (2014). Process and pattern in institutional reforms: A case study of the police pacifying units (UPPs) in Brazil. *World Development*, 64, 232–242.
- Zenou, Y., & Boccoard, N. (2000). Racial discrimination and redlining in cities. *Journal of Urban Economics*, 48(2), 260–285.

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Pongeluppe, L. S. (2022). The favela effect: Spatial inequalities and firm strategies in disadvantaged urban communities. *Strategic Management Journal*, 1–32. <https://doi.org/10.1002/smj.3414>

ONLINE APPENDIX FOR

THE FAVELA EFFECT:

SPATIAL INEQUALITIES AND FIRM STRATEGIES IN DISADVANTAGED URBAN COMMUNITIES

LEANDRO S. PONGELUPPE¹

Rotman School of Management at the University of Toronto

E-mail: L.Pongeluppe@rotman.utoronto.ca

ORCID: <https://orcid.org/0000-0001-6195-4455>

¹ Rotman School of Management, University of Toronto. 105 St George Street, Toronto, ON M5S 3E6, Canada.

ONLINE APPENDIX

CONTENT

ONLINE APPENDIX 1 – CHARACTERISTICS OF FAVELAS AND THEIR SURROUNDINGS IN THE SAMPLE.....	3
ONLINE APPENDIX 2 – FIRM AND PRODUCT CATEGORY HETEROGENEITY TESTS	7
ONLINE APPENDIX 3 – ALTERNATIVE MEASURES TESTS AND INTEGRATION TESTS	13
ONLINE APPENDIX 4 – HECKMAN TEST	25
ONLINE APPENDIX 5 – PREDICTION MODEL OF DELIVERY PRICING FOR FAVELAS	27
ONLINE APPENDIX 6 – E-COMMERCE FIRMS CHARACTERISTICS AND COMPARISON.....	33
ONLINE APPENDIX 7 – VISUAL EVIDENCE FROM THE FIELD.....	35
ONLINE APPENDIX 8 – SLOPE SIGNIFICANCE ANALYSIS	36
ONLINE APPENDIX 9 – MCCRARY TEST AND BANDWIDTH ANALYSIS	39
ONLINE APPENDIX 10 – INFRASTRUCTURE ANALYSIS.....	44
ONLINE APPENDIX 11 – CRIMINALITY ANALYSIS	46
ONLINE APPENDIX 12 – OLS TEST ON THE BINARY OUTCOMES.....	50
ONLINE APPENDIX 13 – FAVELA CONSUMERS’ PURCHASE AND PURCHASE CHANNELS	51
ONLINE APPENDIX 14 – DISTANCE DATA COLLECTION ILLUSTRATION.....	53
ONLINE APPENDIX REFERENCES.....	54

ONLINE APPENDIX 1 – CHARACTERISTICS OF FAVELAS AND THEIR SURROUNDINGS IN THE SAMPLE

City/State	Location Name	Favela or Surrounding	Population (2010)	Pop. Density (people per km ²)	HDI	Avg. Income per capita (R\$)	Poverty vulnerability (%)	Child Mortality (per 1000)	Illiteracy (%)	Labor Informality (%)	Graduate (%)	Garbage collection coverage (%)	Water and sewage coverage (%)	Electricity coverage (%)
Rio de Janeiro	Rocinha	Favela	69,356	53,365	0.663	R\$ 525.68	27.1%	17.6	9.3%	30.7%	2.5%	98.5%	98.3%	100.0%
	Gavea/Sao Conrado	Surrounding	12,352	8,966	0.946	R\$ 6,522.52	0.6%	6.3	0.4%	2.6%	68.3%	100.0%	98.6%	100.0%
	Complexo do Alemao	Favela	58,962	26,823	0.656	R\$ 411.41	35.0%	16.6	9.6%	35.7%	1.5%	92.2%	96.6%	100.0%
	Inhauma	Surrounding	20,659	10,144	0.789	R\$ 943.52	8.9%	11.2	1.9%	22.3%	14.7%	99.6%	99.2%	100.0%
	Jacarezinho	Favela	35,828	37,219	0.670	R\$ 438.22	35.1%	16.9	6.1%	30.4%	1.0%	94.6%	99.0%	100.0%
	Maria da Graça	Surrounding	11,053	11,264	0.844	R\$ 1,359.45	7.2%	9.7	1.1%	16.3%	21.0%	100.0%	98.5%	100.0%
	Complexo da Mare Ramos / Bonsucesso	Favela	29,078	23,198	0.686	R\$ 527.58	26.6%	15.6	8.1%	31.1%	1.9%	99.7%	99.2%	100.0%
		Surrounding	8,974	18,274	0.837	R\$ 1,196.91	8.2%	9.6	1.5%	16.4%	15.6%	100.0%	97.2%	100.0%
	Rio das Pedras Anil / Jacarepagua	Favela	26,098	95,160	0.621	R\$ 524.73	22.5%	17.6	16.0%	33.0%	0.4%	93.1%	98.4%	100.0%
		Surrounding	33,196	1,923	0.903	R\$ 2,360.96	3.9%	7.7	1.5%	9.3%	39.5%	99.9%	99.8%	100.0%
	Vila Vintem	Favela	10,253	31,845	0.691	R\$ 500.89	29.4%	18.3	5.5%	33.7%	3.2%	99.3%	99.5%	100.0%
	Padre Miguel	Surrounding	16,428	13,028	0.796	R\$ 907.10	12.5%	11.2	1.8%	20.4%	12.0%	100.0%	98.6%	100.0%
	Acari	Favela	9,839	9,802	0.657	R\$ 420.62	36.6%	21.6	10.1%	34.9%	1.8%	99.7%	92.6%	100.0%
	Coelho Neto	Surrounding	3,819	8,457	0.782	R\$ 830.00	11.6%	11.8	2.0%	20.0%	9.2%	99.9%	90.7%	100.0%
	Guararapes	Favela	2,097	37,722	0.684	R\$ 580.41	21.5%	16.2	7.7%	27.1%	1.9%	99.9%	97.8%	99.9%
	Cosme Velho	Surrounding	5,081	5,785	0.946	R\$ 4,845.64	1.2%	6.6	1.8%	3.7%	66.4%	100.0%	99.8%	100.0%
Rio de Janeiro City			11,945,976	1,773	0.771	R\$ 1,130.75	21.1%	13.7	4.2%	31.9%	15.6%	97.2%	95.1%	100.0%
Rio de Janeiro State			15,989,929	365	0.761	R\$ 1,039.30	22.3%	14.15	5.1%	33.1%	14.3%	97.6%	95.8%	99.9%

City/State	Location Name	Favela or Surrounding	Population (2010)	Pop. Density (people per km²)	HDI	Avg. Income per capita (R\$)		Poverty vulnerab ility (%)	Child Mortality (per 1000)	Illiteracy (%)	Labor Informality (%)	Graduate (%)	Garbage collection coverage (%)	Water and sewage coverage (%)	Electricity coverage (%)
Brasília	Sol Nascente Ceilandia Sul	Favela Surrounding	56,387	2,815	0.663	R\$	440.54	37.3%	23.5	7.0%	32.5%	2.0%	74.9%	92.3%	100.0%
			28,900	10,061	0.801	R\$	919.90	12.8%	13.5	3.9%	18.2%	9.8%	100.0%	98.0%	100.0%
	Brasília, Federal District		2,570,160	441	0.824	R\$	1,715.11	16.0%	14.01	4.3%	28.4%	24.0%	98.9%	98.1%	99.9%
	Goias State		6,003,788	18	0.735	R\$	810.97	24.2%	13.96	10.1%	43.2%	10.3%	98.7%	96.8%	99.4%
São Paulo	Paraisopolis	Favela	56,369	47,856	0.639	R\$	469.78	29.2%	20.8	13.2%	31.2%	1.6%	100.0%	96.2%	98.8%
	Vila Suzana	Surrounding	19,490	10,846	0.893	R\$	3,109.22	5.3%	7.3	0.8%	13.3%	41.1%	100.0%	97.1%	100.0%
	Heliopolis Sao Joao Climaco	Favela Surrounding	43,747	44,993	0.696	R\$	537.61	25.3%	17.1	7.8%	31.0%	2.0%	100.0%	99.9%	100.0%
			5,114	14,197	0.896	R\$	2,039.78	2.9%	8	0.9%	10.0%	38.6%	100.0%	99.2%	100.0%
	São Paulo City		19,683,975	2,477	0.794	R\$	1,249.72	16.6%	13	4.3%	26.8%	17.1%	99.6%	96.8%	99.6%
	São Paulo State		41,262,199	266	0.783	R\$	1,084.46	16.1%	13.86	5.2%	28.6%	15.1%	99.6%	98.6%	99.9%
São Luís	Coroadinho Coroado	Favela Surrounding	49,338	9,635	0.664	R\$	305.58	55.5%	31.1	11.1%	33.0%	1.3%	92.4%	59.7%	99.9%
			682	3,436	0.897	R\$	2,214.59	10.4%	8.6	1.0%	7.8%	43.2%	99.1%	97.6%	100.0%
	São Luís City		1,331,181	459	0.755	R\$	710.73	38.8%	19.3	7.0%	45.3%	11.9%	90.6%	81.2%	99.9%
	Maranhão State		6,574,789	20	0.639	R\$	360.34	63.6%	28.03	27.2%	66.8%	5.4%	79.1%	82.1%	96.1%
Fortaleza	Pirambu Alvaro Weyne	Favela Surrounding	35,314	28,325	0.678	R\$	392.26	41.7%	18.3	12.4%	36.0%	2.6%	99.0%	91.7%	99.2%
			21,810	14,873	0.764	R\$	672.11	30.3%	14.23	5.2%	25.5%	9.0%	99.5%	96.3%	99.7%
	Fortaleza City		3,615,767	624	0.732	R\$	688.72	39.7%	16.78	11.7%	43.3%	10.7%	96.0%	91.6%	99.6%
	Ceará State		8,452,381	57	0.682	R\$	460.63	54.9%	19.29	24.0%	58.4%	7.2%	93.6%	86.1%	99.1%

City/State	Location Name	Favela or Surrounding	Population (2010)	Pop. Density (people per km²)	HDI	Avg. Income per capita (R\$)	Poverty vulnerability (%)	Child Mortality (per 1000)	Illiteracy (%)	Labor Informality (%)	Graduate (%)	Garbage collection coverage (%)	Water and sewage coverage (%)	Electricity coverage (%)
Belém	Paar	Favela	31,658	13,546	0.696	R\$ 413.46	46.5%	21.5	4.6%	28.4%	3.3%	99.7%	86.9%	100.0%
	Cidade Nova	Surrounding	38,752	11,749	0.824	R\$ 905.01	11.8%	11.7	1.3%	13.7%	15.4%	100.0%	97.5%	100.0%
	Bengui	Favela	29,379	15,264	0.676	R\$ 430.61	40.4%	22.2	5.9%	33.8%	2.6%	99.1%	89.5%	100.0%
	Parque Verde	Surrounding	1,086	2,896	0.881	R\$ 1,709.15	7.1%	9.8	1.2%	9.4%	37.5%	99.4%	95.3%	100.0%
	Condor	Favela	2,903	45,449	0.573	R\$ 264.24	63.8%	31.4	5.8%	49.9%	1.7%	87.5%	52.0%	93.0%
	Jurunas	Surrounding	49,572	23,403	0.674	R\$ 433.34	39.5%	21.8	6.0%	34.1%	4.0%	99.2%	90.2%	99.9%
	Belém City		2,275,032	638	0.729	R\$ 715.11	37.1%	19.1	4.9%	47.2%	11.3%	96.7%	85.6%	99.7%
	Pará State		7,581,051	6	0.646	R\$ 446.76	56.0%	20.29	15.0%	62.6%	6.2%	91.9%	84.7%	91.9%
Recife	Casa Amarela	Favela	31,426	18,787	0.664	R\$ 360.89	47.4%	23.8	11.3%	36.3%	3.1%	95.4%	90.9%	99.6%
	Cajueiro	Surrounding	4,486	10,831	0.860	R\$ 1,341.98	13.9%	9.4	2.4%	11.0%	31.1%	98.9%	96.7%	100.0%
	Recife City		3,690,547	1,331	0.734	R\$ 780.36	39.5%	17.3	10.5%	37.2%	12.2%	95.4%	92.0%	99.9%
	Pernambuco State		8,796,448	90	0.673	R\$ 525.64	51.9%	20.43	22.2%	53.5%	8.0%	94.7%	83.7%	99.5%
Curitiba	Cajuru	Favela	29,926	6,641	0.665	R\$ 501.43	25.5%	17.7	8.6%	34.0%	2.2%	99.2%	95.8%	100.0%
	Centenario/Mercurio	Surrounding	9,444	5,905	0.841	R\$ 1,397.90	5.9%	7.87	1.3%	15.5%	24.0%	100.0%	99.6%	100.0%
	Curitiba City		3,223,836	194	0.783	R\$ 1,183.32	13.1%	11.2	4.2%	27.7%	17.8%	99.7%	98.3%	99.9%
	Paraná State		10,444,526	52	0.749	R\$ 890.89	19.7%	13.08	7.9%	34.4%	12.8%	99.2%	96.7%	99.6%
Manaus	Cidade de Deus	Favela	25,209	9,946	0.637	R\$ 360.03	49.0%	25.7	6.5%	35.2%	2.1%	97.1%	77.6%	99.6%
	Cidade Nova	Surrounding	7,550	9,433	0.762	R\$ 723.45	22.8%	15	1.9%	19.9%	10.8%	98.6%	96.3%	100.0%
	Manaus City		2,106,322	21	0.720	R\$ 725.17	37.4%	17.39	6.2%	42.9%	10.1%	98.0%	86.7%	98.5%
	Amazonas State		3,483,985	2	0.674	R\$ 539.80	51.8%	17.01	12.3%	55.4%	8.2%	93.0%	77.3%	92.2%

City/State	Location Name	Favela or Surrounding	Population (2010)	Pop. Density (people per km²)	HDI	Avg. Income per capita (R\$)		Poverty vulnerab ility (%)	Child Mortality (per 1000)	Illiteracy (%)	Labor Informality (%)	Graduate (%)	Garbage collection coverage (%)	Water and sewage coverage (%)	Electricity coverage (%)
Porto Alegre	Cruzeiro do Sul Jardim Medianeira	Favela Surrounding	12,523	23,005	0.627	R\$	531.90	30.0%	17.2	9.0%	35.6%	1.8%	100.0%	95.1%	100.0%
			7,031	7,576	0.853	R\$	1,669.56	2.5%	7.2	1.1%	13.6%	26.3%	100.0%	99.8%	100.0%
	Porto Alegre City		4,031,688	390	0.762	R\$	1,143.12	15.5%	11.1	3.7%	27.2%	14.0%	99.5%	97.6%	99.9%
	Rio Grande do Sul State		10,693,929	38	0.746	R\$	959.24	18.7%	12.38	5.4%	33.6%	11.3%	99.2%	96.2%	99.7%
Salvador	Alto das Pombas Barra	Favela Surrounding	10,152	30,120	0.690	R\$	492.16	32.3%	19.05	7.1%	25.4%	3.2%	93.5%	87.9%	99.6%
			16,150	12,143	0.926	R\$	3,531.90	3.5%	7.6	1.0%	5.2%	53.7%	99.5%	99.0%	100.0%
	Salvador City		3,573,973	821	0.743	R\$	874.07	33.3%	16.2	5.9%	34.4%	12.6%	95.9%	94.6%	99.8%
	Bahia State		14,016,906	25	0.660	R\$	496.73	52.7%	21.73	20.9%	57.1%	6.4%	95.4%	85.7%	96.5%

Note: Brazilian National 2010 Census Data at *Atlas do Desenvolvimento Humano no Brasil*. Available at: <http://atlasbrasil.org.br/2013/pt/> Accessed on March 19th, 2020. The geo-referencing system of the Brazilian census split some of the considered areas. Therefore, for the purposes of illustration, I selected the most representative units to perform the comparison. I included all the biggest favelas of all the major state capitals in the country to avoid possible regional biases. The two biggest cities in the country, São Paulo and Rio de Janeiro, have in total ten favelas included to avoid misrepresentation bias of these locations in the final sample. Also, the city of Belém has three favelas included because it is the state capital with the highest percentage of inhabitants living in favelas (54.5% of the total city population).

ONLINE APPENDIX 2 – FIRM AND PRODUCT CATEGORY HETEROGENEITY TESTS

Firm Heterogeneity

Firm heterogeneity analyses help to pin down whether firms are indeed heterogeneous and consequently might be adopting different strategies to transact with favela consumers. I start evaluating this potential heterogeneity through an interaction term of the favela dummy with each firm in my sample (see Table 1). According to the LOGIT model (2), and the OLS models (2) and (4), e-commerce firms are heterogeneous in their strategies toward favela customers. The analyses present a joint-significance χ^2 test equals to 13.80 ($p = 0.032$), which confirms that firms are heterogeneous in their likelihood to refuse orders from favelas residents. LOGIT model (2) shows that some firms are less likely to refuse an order from favela consumers, for example, firms #03 and #07, while others seem to be more likely to do so, such as firm #10. Moreover, firms #02, #08, and #09 never refuse any order neither to favela nor non-favela consumers. On the other hand, firm #05 only refuses orders requested by favela consumers, which explains why these firms are omitted in the main specification.

Also, the OLS model (2) and (4) present a joint-significance F -test equals 2.26 ($p = 0.058$), and F -test equals 2.43 ($p = 0.044$), respectively, for delivery price and delivery price ratio, which confirm that firms are heterogeneous both measures toward favela consumers.² The models show that e-commerce firms #03, #04, and #09 and have statistically lower delivery prices and/or delivery price ratios towards favela consumers. In other words, some firms seem to develop strategies to reduce the delivery price ratio from 50% (e.g., firm #03) to almost 100% (e.g., firm #09) of the average delivery price ratio used in the industry toward favelas consumers.

² Notice that the p-value of F-test for OLS model (2) is ($p < 0.058$) which I am considering as adequate for the purposes of the study analyzes.

Nevertheless, firm #06 and #08 have a statistically significant higher delivery price and delivery price ratios toward favelas consumers.

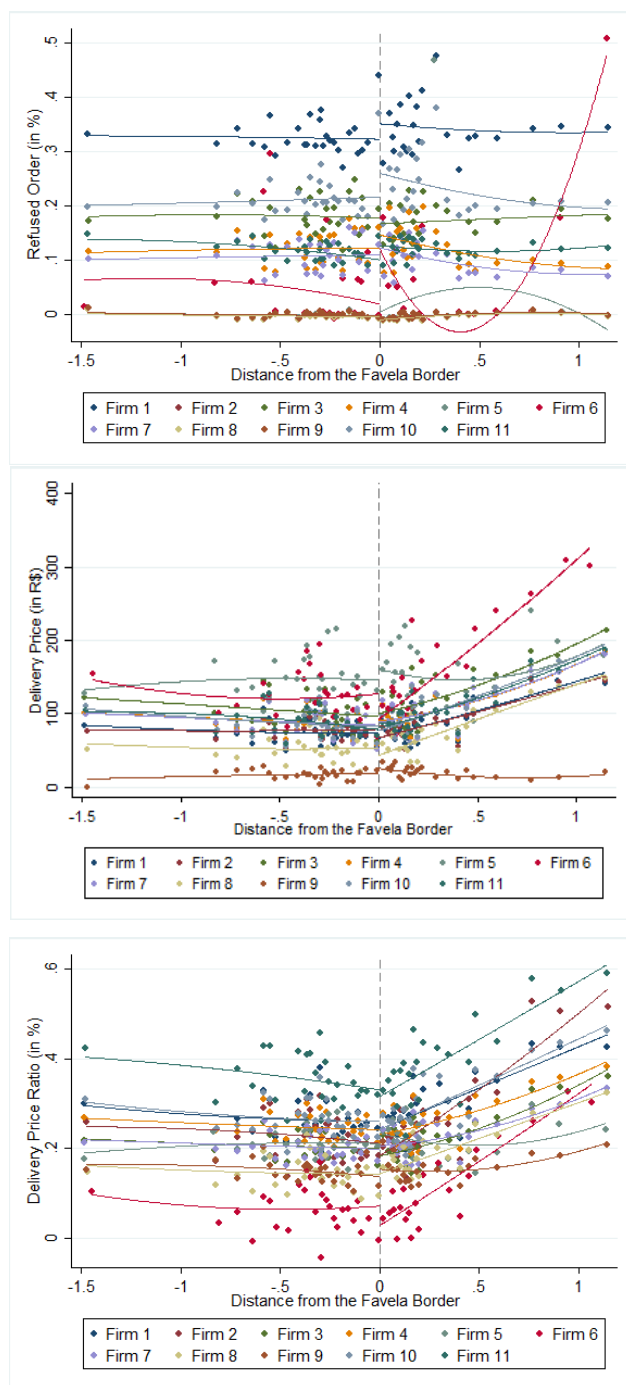


Figure 1 – Non-parametric Estimation of Refused Order (in %) Delivery Price (in R\$), and Delivery Price Ratio (in %), by firm

Note: A non-parametric estimation was performed considering 50 bins and a discontinuity point at the favela border. All negative distances represent zip codes outside the favelas and all positive distances zip codes inside the favelas. The graphs were elaborated controlling for firm, week, location, and product category fixed effects. Note that the y-axes of the graphs are on different scales.

Table 1 – Firm Heterogeneity on the “Favela Effect”

	LOGIT (1) 1(Refused Order)	LOGIT (2)	OLS (1) Delivery Price (in R\$)	OLS (2)	OLS (3) Delivery Price Ratio (in %)	OLS (4)
Favela	0.109* (0.0531)	0.198** (0.0244)	14.24* (5.079)	18.83** (1.039)	0.0344** (0.0117)	0.0308** (0.00184)
<i>Firm Heterogeneous Effect</i>						
Favela × Firm #02			-5.016* (2.089)	-4.065* (1.730)	0.00815 (0.00891)	0.0162+ (0.00835)
Favela × Firm #03	-0.163* (0.0670)	-0.201* (0.0823)	3.888 (3.176)	3.172 (2.747)	-0.0211* (0.00981)	-0.0129* (0.00478)
Favela × Firm #04	-0.0615 (0.0666)	-0.0999 (0.0681)	3.268 (2.080)	0.400 (2.008)	-0.0209* (0.00932)	-0.0115** (0.00335)
Favela × Firm #05			-9.644 (10.06)	-8.566 (9.649)	-0.0297+ (0.0157)	-0.0214 (0.0144)
Favela × Firm #06	0.182 (0.273)	0.154 (0.295)	8.778+ (4.577)	10.41* (4.254)	-0.000555 (0.00630)	0.00910* (0.00425)
Favela × Firm #07	-0.156** (0.0605)	-0.202** (0.0709)	2.345 (2.177)	0.779 (1.866)	-0.0142* (0.00626)	-0.00486 (0.00338)
Favela × Firm #08			2.535 (1.979)	3.459* (1.514)	0.00511 (0.00806)	0.0131* (0.00494)
Favela × Firm #09			-13.66** (4.786)	-12.69** (4.450)	-0.0325** (0.00849)	-0.0247** (0.00735)
Favela × Firm #10	0.0702+ (0.0394)	0.0703+ (0.0400)	4.107+ (2.122)	1.719 (2.075)	0.00426 (0.00348)	0.00344 (0.00285)
Favela × Firm #11	-0.0478 (0.0662)	-0.0785 (0.0803)	0.473 (2.920)	1.896 (2.976)	0.00103 (0.00781)	0.00801 (0.00638)
Week Fixed Effect	N	Y	N	Y	N	Y
Product Category Fixed Effect	N	Y	N	Y	N	Y
Firm Fixed Effect	Y	Y	Y	Y	Y	Y
Location Fixed Effect	N	Y	N	Y	N	Y
Constant	-0.735** (0.0543)	0.488** (0.122)	70.46** (10.94)	19.11* (8.975)	0.246** (0.0267)	0.0610** (0.0181)
Observations	286,148	286,148	354,077	354,077	354,077	354,077
Pseudo and R-squared	0.0533	0.144	0.036	0.334	0.048	0.386
Log-Likelihood	-126,045	-113,939				
chi2 (Favela × Firm)	13.67	13.80				
chi2 (Favela × Firm) p-value	0.034	0.032				
F-test (Favela × Firm)			3.47	2.26	2.33	2.43
F-test (Favela × Firm) p-value			0.009	0.058	0.051	0.044

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. The models used Firm #01 as the baseline (omitted). Refused Order models omit firms that never refuse (Firms #02, #08, and #09) and firms that when they refuse, it is always to customers in favelas (firm #05). ** p<0.01, * p<0.05, + p<0.1

Product Category Heterogeneity

The *transaction costs* literature predicts that product characteristics explain heterogeneity among firms (North, 1992) because a product's value and size shape delivery costs that may arise from robbery risks and/or logistical challenges (Duarte, Macau, Flores e Silva, & Sanches, 2019; Goldfarb & Tucker, 2017).

To test for this rationale, I use the **product category variable**, representing four categories, which vary by size (small vs. large) and value (cheap vs. expensive).³ These categories are justified by the findings of specialized literature on logistical distribution in slum markets (Anderson, Markides, & Kupp, 2010; Duarte et al., 2019; Meirelles & Athayde, 2016; Perlman, 2010). Theoretical predictions from *transaction costs* suggest that firms would charge more on “expensive and large” products because they are more affected by potential higher transaction costs of favelas. Similarly, “cheap and small” products are expected to be less affected. Online Appendix 5 describes this rationale mathematically.

³ More specifically, the products are the following: (i) Expensive and Small: cellphones and laptops; (ii) Expensive and Large: televisions, and refrigerators or freezers; (iii) Cheap and Small: books, games or DVDs, t-shirts, makeup or bags; and (iv) Cheap and Large: sofas, and beds.

Table 2 presents the results for the product category heterogeneity test. The joint-significance tests on coefficients cannot reject the null hypothesis on both the absolute and proportional delivery charges. Respectively, the *F-tests* are equal to 2.30 ($p = 0.108$), and 2.14 ($p = 0.127$). There is no statistically significant heterogeneity on the delivery charge by product category. Moreover, in the LOGIT model, despite rejecting the null in the refused order variable for which the *chi*² test is equal to 8.76 ($p = 0.033$), the results show that orders in favelas for “cheap and large” products are less likely than others to be refused ($p < 0.01$). This result is not consistent with the predictions of the transaction costs theory about large-sized products for which the potential logistical difficulties would drive a higher, not lower, likelihood of refusal.⁴

Despite the theoretical predictions about expensive and large-sized products, the quantitative results reject the argument that product characteristics are the source of heterogeneity. *Transaction costs* differences do not explain the findings.

⁴ Table 4 in the Online Appendix 3 is a robustness check of these results using the two alternative measures. The results follow a similar pattern in which “Out of Stock” *chi*² test equals to 16.72 ($p = 0.001$), and delivery time *F-test* equals to 1.64 ($p = 0.212$). Curiously, in these tests both “cheap” and “expensive” large-sized products have a lower likelihood of displaying the “Out of Stock” message to favelas consumers ($p < 0.01$). Again, this result goes in the opposite direction of what the theoretical predictions would suggest. Therefore, the robustness check helps me to reject transaction costs explanation.

Table 2 – Product Category Heterogeneity Analysis

	LOGIT (1) 1(Refused Order)	OLS (1) Delivery Price (in R\$)	OLS (2) Delivery Price Ratio (in %)
Favela × Expensive and Small	-	-	-
Favela × Expensive and Large	-0.0451 (0.0581)	20.84* (8.223)	0.0129* (0.00521)
Favela × Cheap and Small	-0.0738 (0.0581)	-2.027* (0.788)	0.0280* (0.0124)
Favela × Cheap and Large	-0.388** (0.138)	31.70* (12.75)	0.0560* (0.0228)
Favela	0.256** (0.0469)	8.670* (4.009)	0.00537 (0.0105)
Week Fixed Effect	Y	Y	Y
Product Category Fixed Effect	Y	Y	Y
Firm Fixed Effect	Y	Y	Y
Location Fixed Effect	Y	Y	Y
Constant	0.460** (0.127)	24.01* (8.440)	0.0730** (0.0163)
Observations	288,679	354,077	354,077
Pseudo and R-squared	0.147	0.336	0.386
Log-Likelihood	-113,953		
chi2 (Favela × Product Category)	8.76		
chi2 (Favela × Product Category) p-value	0.033		
F-test (Favela × Product Category)		2.30	2.14
F-test (Favela × Product Category) p-value		0.108	0.127

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. The models used “Expensive and Small” product category as the baseline (omitted). *** p<0.01, ** p<0.05, * p<0.1

ONLINE APPENDIX 3 – ALTERNATIVE MEASURES TESTS AND INTEGRATION TESTS

To check the robustness of the results presented in the main paper, I evaluate whether the results hold using alternative measures of my dependent variables. In particular, I estimate whether the results hold for two alternative variables: **Out of stock** (dummy), which is the most used justification by firms refusing orders from favela consumers. And, **Delivery time** (in days), which measures the total number of days estimated to perform the delivery.

Table 3 shows the average “favela effect” now evaluated using these two alternative measures of the dependent variable. The LOGIT model (2) shows that firms are more likely to display the “Out of Stock” message to favela consumers ($p < 0.01$). Also, the OLS model (2) suggests that favela consumers receive their products one day later on average ($p < 0.01$). Both results are consistent with findings for the main effect. Figure 2 shows these results in both a parametric and a non-parametric plot. Similarly, Table 4 LOGIT model (2) and OLS model (2) show that firms are jointly heterogeneous both on their “Out of Stock” justification likelihood, and on their delivery time strategies, respectively χ^2 test equals 13.11 ($p = 0.011$), and delivery time F -test equals 2.77 ($p = 0.025$). Thus, in support of previous analysis, we can see that firms are heterogeneous concerning these alternative dependent variables as well.

Table 5 brings a robustness check for *transaction costs rationale*: product category heterogeneity – using the two alternative measures. The results follow a similar pattern to the main variables. In the present case, “Out of Stock” χ^2 test equals to 16.72 ($p = 0.001$), and delivery time F -test equals to 1.64 ($p = 0.212$). Curiously, in these tests, both ‘cheap and large’ and “expensive and large” products have a lower likelihood to display the “Out of Stock” message to favelas consumers at ($p < 0.01$). Therefore, the robustness check helps me to reject the transaction costs explanation.

Table 6 brings a robustness check for the *resources and capabilities* rationale: physical stores as complementary assets – using the two alternative measures. The results follow a similar pattern to the main variables. In the present analyses, physical stores, either inside or outside favelas, reduce the likelihood of an “Out of Stock” message to favelas consumers ($p < 0.05$). Moreover, the results show that firms with physical stores inside favelas have a faster delivery time than their competitors, which do not have stores in favelas ($p < 0.01$). Therefore, the robustness check helps me to support the resources and capabilities explanations on the importance of complementary assets.

Table 7 brings a robustness check for the *stakeholder management* rationale: employees and consumers’ assessment of the firm – using the two alternative measures. The results follow a similar pattern to the main variables. In the present case, firms with better employees’ assessment and higher responsiveness to consumer complaints reduce the likelihood of an “Out of Stock” message to favelas consumers ($p < 0.05$). However, the results show that firms with responsiveness scores with consumers have a slower delivery time in favelas than their competitors at ($p < 0.05$). The robustness check lends validity to stakeholder management rationale, suggesting that firm stakeholder orientation might shift a superior treatment of favelas consumers, with the caveat that this might increase the delivery time. Following the main paper, Figure 3 illustrates results comparing top-three and not top-three firms in both a parametric and a non-parametric plot.

Table 8 and Table 9 bring the *integration test* between *resources and capabilities* and *stakeholder management* rationales using Firm #03 as an illustrative case. The results reinforce the conclusion that combining physical stores with stakeholder orientation lead firms to a more equitable treatment of favela consumer. The results are robust considering both the main model and the robustness checks.

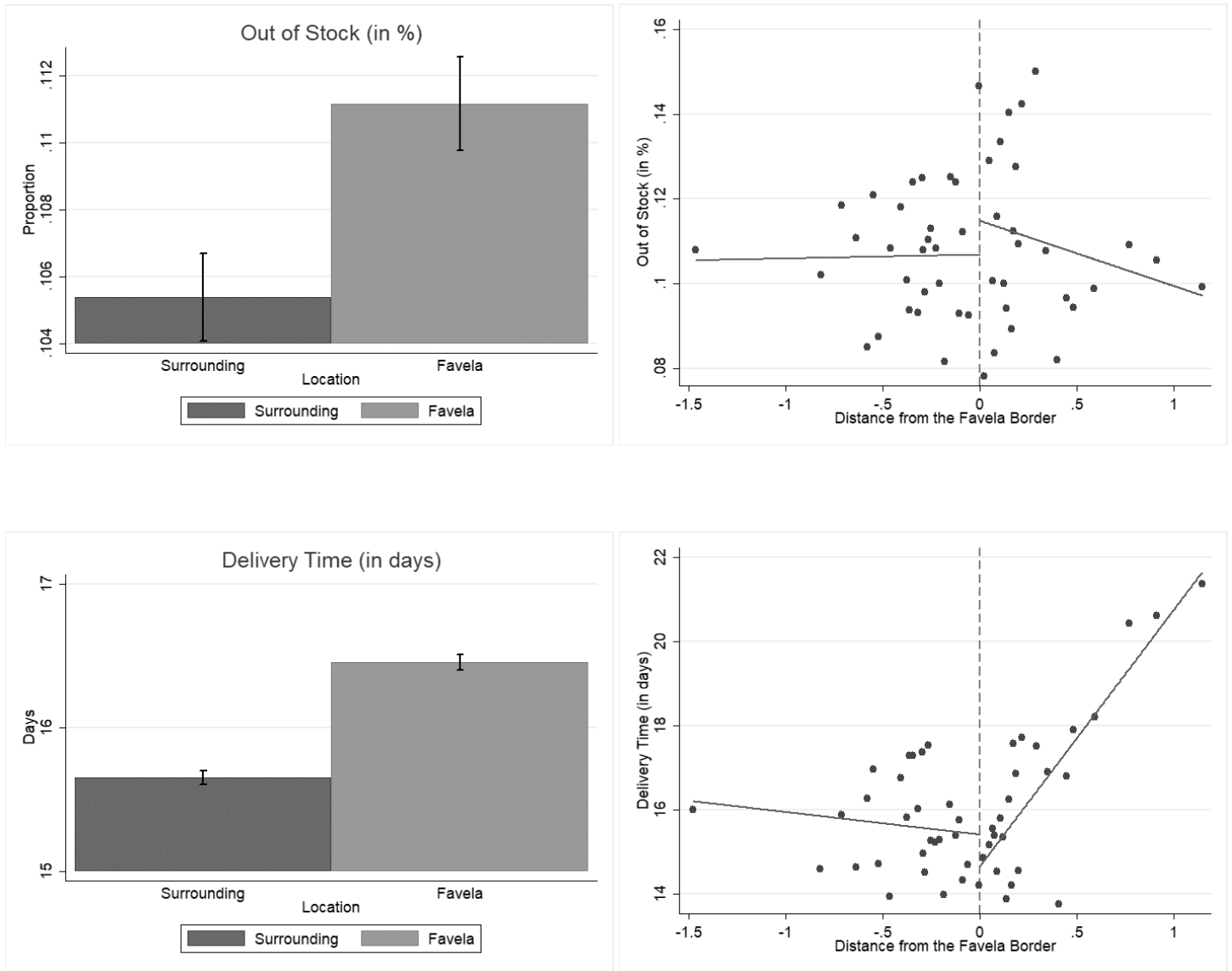


Figure 2 – Parametric and Non-parametric Estimation of “Out of Stock” (in %) Delivery Time (in days), Aggregated

Note: A non-parametric estimation was performed considering 50 bins and a discontinuity point at the favela border. All negative distances represent zip codes outside the favelas and all positive distances zip codes inside the favelas. The graphs were elaborated controlling for firm, week, location, and product category fixed effects. Note that the y-axes of the graphs are on different scales.

Table 3 – The “Favela Effect,” Alternative Measures

	LOGIT (1)	LOGIT (2)	OLS (1)	OLS (2)
	1(Out of Stock)		Delivery Time (in Days)	
Favela	0.0599 (0.0447)	0.0337** (0.000486)	0.804* (0.344)	0.892** (0.00597)
Week Fixed Effect	N	Y	N	Y
Product Category Fixed Effect	N	Y	N	Y
Firm Fixed Effect	N	Y	N	Y
Location Fixed Effect	N	Y	N	Y
Constant	-2.139** (0.0705)	0.877** (0.114)	15.65** (0.822)	10.63** (0.535)
Observations	406,900	225,684	354,077	354,077
Pseudo and R-squared	0.000	0.166	0.001	0.471
Log-Likelihood	-139,394	-92,852		

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil.

** p<0.01, * p<0.05, + p<0.1

Table 4 – Firm Heterogeneity on the “Favela Effect,” Alternative Measures

	LOGIT (1)	LOGIT (2)	OLS (1)	OLS (2)
	1(Out of Stock)		Delivery Time (in Days)	
Favela	0.109*	0.0953**	0.747+	0.790**
	(0.0531)	(0.0236)	(0.364)	(0.122)
<i>Firm Heterogeneous Effect</i>				
Favela × Firm #02			-0.176	-0.106
			(0.234)	(0.228)
Favela × Firm #03	-0.163*	-0.215*	0.144	0.107
	(0.0670)	(0.0881)	(0.322)	(0.312)
Favela × Firm #04	-0.0615	-0.114+	0.230	0.131
	(0.0666)	(0.0687)	(0.281)	(0.289)
Favela × Firm #05			-0.371	-0.298
			(0.321)	(0.312)
Favela × Firm #06			-0.300	-0.167
			(0.274)	(0.283)
Favela × Firm #07	-0.156**	-0.218**	0.331	0.278
	(0.0605)	(0.0751)	(0.280)	(0.280)
Favela × Firm #08			-0.544*	-0.478*
			(0.241)	(0.227)
Favela × Firm #09			0.552	0.619*
			(0.323)	(0.296)
Favela × Firm #10	0.0702+	0.0732+	0.334+	0.236+
	(0.0394)	(0.0418)	(0.185)	(0.134)
Favela × Firm #11			0.306+	0.388*
			(0.160)	(0.156)
Week Fixed Effect	N	Y	N	Y
Product Category Fixed Effect	N	Y	N	Y
Firm Fixed Effect	Y	Y	Y	Y
Location Fixed Effect	N	Y	N	Y
Constant	-0.735**	0.849**	17.73**	10.68**
	(0.0543)	(0.120)	(0.772)	(0.539)
Observations	225,684	225,684	354,077	354,077
Pseudo and R-squared	0.0476	0.167	0.214	0.471
Log-Likelihood	-106,048	-92,804		
chi2 (Favela × Firm)	12.86	13.11		
<i>chi2 (Favela × Firm) p-value</i>	<i>0.012</i>	<i>0.011</i>		
F-test (Favela × Firm)			3.00	2.77
<i>F-test (Favela × Firm) p-value</i>			<i>0.018</i>	<i>0.025</i>

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018.

The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. The models used Firm #01 as a baseline (omitted). ** p<0.01, * p<0.05, + p<0.1

Table 5 – Product Category Heterogeneity Analysis, Alternative Measures

	LOGIT (1) 1(Out of Stock)	OLS (1) Delivery Time (in Days)
Favela × Expensive and Small	-	-
Favela × Expensive and Large	-0.0806** (0.0245)	0.572+ (0.277)
Favela × Cheap and Small	-0.0894 (0.0703)	0.177 (0.206)
Favela × Cheap and Large	-0.503** (0.158)	0.411 (0.261)
Favela	0.179** (0.0600)	0.624** (0.172)
Week Fixed Effect	Y	Y
Product Category Fixed Effect	Y	Y
Firm Fixed Effect	Y	Y
Location Fixed Effect	Y	Y
Constant	0.808** (0.131)	10.76** (0.493)
Observations	225,684	354,077
Pseudo and R-squared	0.167	0.471
Log-Likelihood	-92,744	
chi2 (Favela × Product Category)	16.72	
chi2 (Favela × Product Category) p-value	0.001	
F-test (Favela × Product Category)		1.64
F-test (Favela × Product Category) p-value		0.212

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018.

The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. The models used the “Expensive and Small” product category as the baseline (omitted). *** p<0.01, ** p<0.05, * p<0.1

Table 6 – Resources and Capabilities - Physical Stores as Complementary Assets, Alternative Measures

	LOGIT (1) 1(Out of Stock)	LOGIT (2)	OLS (1) Delivery Time (in Days)	OLS (2)
Favela × Firm Has Physical Stores	-0.183* (0.0819)		-0.0841 (0.207)	
Favela × Firm Has Physical Stores in Favelas		-0.184* (0.0825)		-0.383** (0.0950)
Favela	0.172** (0.0621)	0.0684** (0.0151)	0.966** (0.182)	0.975** (0.0196)
Firm Has Physical Stores	0.768** (0.0454)		-0.952** (0.278)	
Firm Has Physical Stores in Favelas		-0.783** (0.104)		-3.772** (0.533)
Week Fixed Effect	Y	Y	Y	Y
Product Category Fixed Effect	Y	Y	Y	Y
Firm Fixed Effect	Y	Y	Y	Y
Location Fixed Effect	Y	Y	Y	Y
Constant	0.131 (0.131)	0.860** (0.119)	11.59** (0.608)	10.60** (0.538)
Observations	225,684	225,684	354,077	354,077
Pseudo and R-squared	0.166	0.166	0.471	0.471
Log-Likelihood	-92830	-92833		

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. ** p<0.01, * p<0.05, + p<0.1



Figure 3 – Graphic Illustration, based on the Employees' Assessment of the Firm Culture (top-three scores), using Alternative Measures “Out of Stock” (in %) Delivery Time (in days)

Note: Parametric and Non-parametric Estimation of “Out of Stock” (in %) Delivery Time (in days). The non-parametric estimation was performed considering 30 bins and a discontinuity point at the favela border. All negative distances represent zip codes outside the favelas and all positive distances zip codes inside the favelas. The graphs were elaborated controlling for firm, week, location, and product category fixed effects. Note that the y-axes of the graphs are on different scales.

Table 7 – Stakeholder Orientation - Employees and Consumers’ Assessment of the Firm, Alternative Measures

	LOGIT (1) 1(Out of Stock)	LOGIT (2)	OLS (1) Delivery Time (in Days)	OLS (2)
Favela × Employees’ Assessment of the Culture	-0.902* (0.363)		0.550 (0.417)	
Favela × Responsiveness to Consumer Complaints		-0.481* (0.193)		0.470* (0.189)
Favela	3.453* (1.378)	47.92* (19.27)	-1.157 (1.553)	-45.92* (18.81)
Employees’ Assessment of the Culture	-6.513** (0.745)		-7.720** (1.749)	
Responsiveness to Consumer Complaints		-3.165** (0.295)		3.506** (0.853)
Week Fixed Effect	Y	Y	Y	Y
Product Category Fixed Effect	Y	Y	Y	Y
Firm Fixed Effect	Y	Y	Y	Y
Location Fixed Effect	Y	Y	Y	Y
Constant	24.94** (2.848)	315.5** (29.38)	39.21** (6.127)	-337.9** (85.12)
Observations	225,684	225,684	354,077	354,077
Pseudo and R-squared	0.166	0.166	0.471	0.471
Log-Likelihood	-92,810	-92,834		

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. ** p<0.01, * p<0.05, + p<0.1

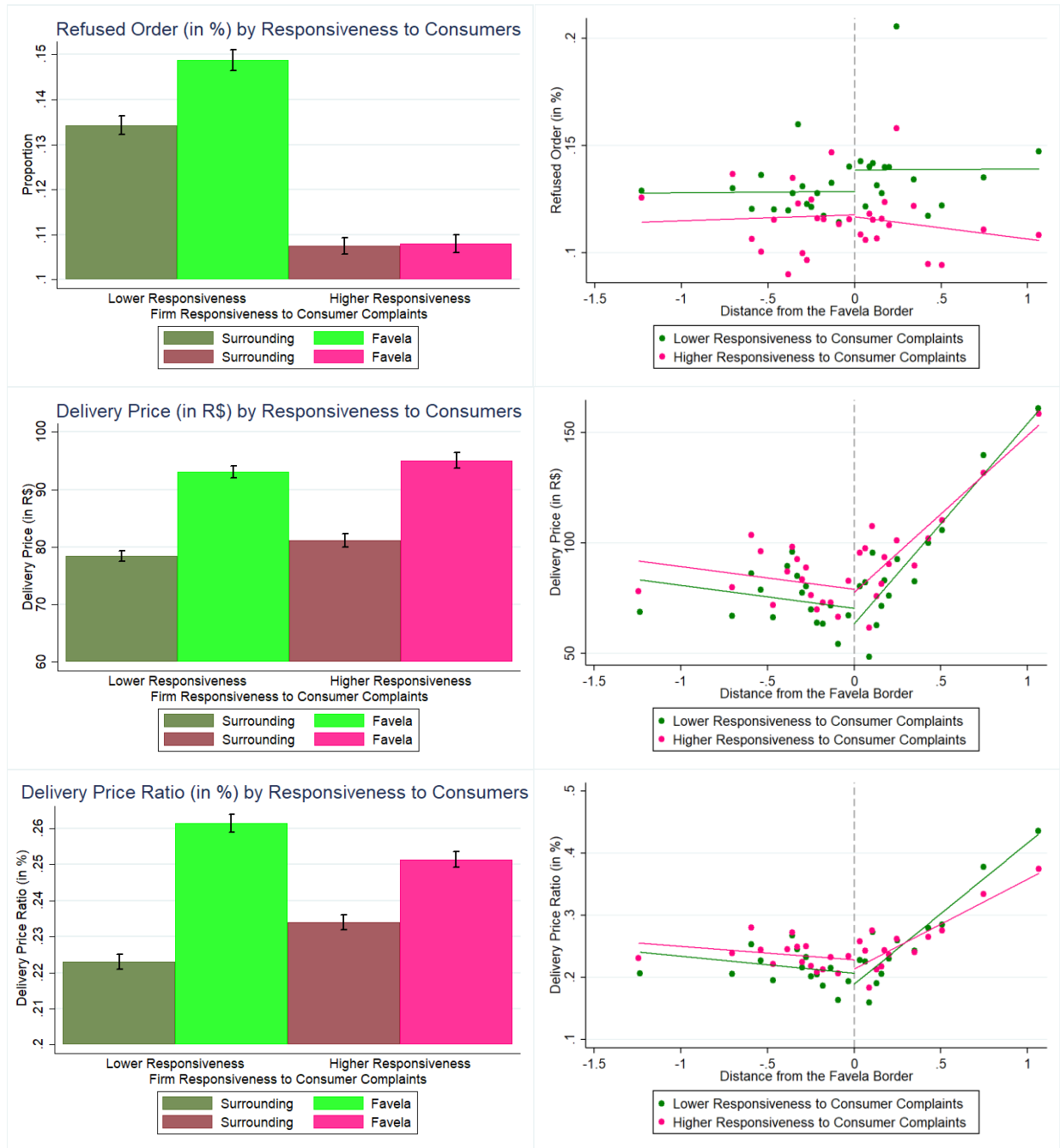


Figure 4 – Graphic Illustration, based on the Firm Responsiveness to Consumer Complaints (Below vs. Above/Equal to the Median)

Note: Parametric and Non-parametric Estimation of Refused Order (in %) Delivery Price (in R\$), and Delivery Price Ratio (in %). The non-parametric estimation was performed considering 30 bins and a discontinuity point at the favela border. All negative distances represent zip codes outside the favelas and all positive distances zip codes inside the favelas. The graphs were elaborated controlling for firm, week, location, and product category fixed effects. Graphs consider median (not top 3 firms) as cut-off given the narrower dispersion of scores in firm responsiveness to consumers. Note that the y-axes of the graphs are on different scales.

Table 8 – Illustration of the Integration of *Resources and Capabilities* with *Stakeholder Management Theories* using Firm #03 Case

	LOGIT (1) 1(Refused Order)	OLS (1) Delivery Price (in R\$)	OLS (2) Delivery Price Ratio (in %)
Favela × Firm #03	-0.175* (0.0702)	3.743 (2.246)	-0.0143* (0.00533)
Favela	0.174** (0.00992)	18.25** (0.268)	0.0321** (0.000555)
Firm #03	-0.774** (0.0959)	11.82* (4.831)	-0.0395** (0.0100)
Week Fixed Effect	Y	Y	Y
Product Category Fixed Effect	Y	Y	Y
Firm Fixed Effect	Y	Y	Y
Location Fixed Effect	Y	Y	Y
Constant	0.499** (0.118)	19.35* (8.984)	0.0603** (0.0184)
Observations	288,679	354,077	354,077
Pseudo and R-squared	0.147	0.334	0.386
Log-Likelihood	-114,013		

Note: standard errors are clustered at 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (size, value), for eleven different e-commerce firms, operating in 21 different favelas and neighborhoods at the main state capitals in Brazil. ** p<0.01, * p<0.05, + p<0.1

Table 9 – Illustration of the Integration of *Resources and Capabilities* with *Stakeholder Management Theories* using Firm #03 Case, Alternative Measures

	LOGIT (1) 1(Out of Stock)	OLS (1) Delivery Time (in Days)
Favela × Firm #03	-0.184* (0.0825)	0.00925 (0.268)
Favela	0.0684** (0.0151)	0.891** (0.0287)
Firm #03	-0.783** (0.104)	-4.577** (0.363)
Week Fixed Effect	Y	Y
Product Category Fixed Effect	Y	Y
Firm Fixed Effect	Y	Y
Location Fixed Effect	Y	Y
Constant	0.860** (0.119)	10.64** (0.531)
Observations	225,684	354,077
Pseudo and R-squared	0.166	0.471
Log-Likelihood	-92833	

Note: standard errors are clustered at 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (size, value), for eleven different e-commerce firms, operating in 21 different favelas and neighborhoods at the main state capitals in Brazil. ** p<0.01, * p<0.05, + p<0.1

ONLINE APPENDIX 4 – HECKMAN TEST

Another concern that might emerge from the previous analysis is the possible selection bias firms insert in the sample. Firms can initially “cherry-pick” the zip codes for which they are likely to accept the order and subsequently set the delivery price in this selected sample. To test for the selectivity problem, I use a Heckman two-step model (Heckman, 1979; Puhani, 2000). The central assumption in the model is that if the coefficient in $\hat{\lambda}$ is significant and not zero, there is evidence that the OLS models estimation are not taking into account firms *ex ante* selection of zip codes.

Table 10 shows the result of the Heckman two-step models for both delivery price and delivery price ratio (main dependent variables), as well as delivery time (alternative dependent variable measure). In all three models, the results show that firms are indeed selecting (“cherry-picking”) zip codes for which they are more likely to transact. And, in the second moment, they define the delivery price, the delivery price ratio, and the delivery time for these selected zip codes. In other words, $\hat{\lambda}$ is significant in all three models ($p < 0.01$).

However, evaluating the coefficient of the favela dummy, the models evidence that the selection bias does not substantially change the $\hat{\beta}_{OLS}$ from my main models. As Table 10 Heckman (1) shows the delivery price is R\$ 19.35 ($p < 0.01$), which is reasonably close to the R\$ 18.65 estimated in the main analysis (Table 3 in the paper). Similarly, in Table 10 Heckman (2), the delivery price ratio is 3.10 p.p. ($p < 0.01$), which is again similar to the 3.06 p.p. in the main analysis (Table 3 in the paper). Finally, in Table 10 Heckman (3), the delivery time is 0.98 days ($p < 0.01$), which is also analogous to the 0.89 days in the main analysis (Table 2 in APPENDIX C). Note that in all three Heckman two-step models, the $\hat{\beta}_{Heckman}$ have a slightly higher magnitude than the $\hat{\beta}_{OLS}$ in the main analysis, nevertheless, the significance level is the same ($p < 0.01$).

Table 10 – Heckman Test on the “Favela Effect”

	HECKMAN (1)	HECKMAN (2)	HECKMAN (3)
	Delivery Price (in R\$)	Delivery Price Ratio (in %)	Delivery Time (in Days)
Favela	19.35** (4.166)	0.0310** (0.00767)	0.980** (0.238)
Week Fixed Effect	Y	Y	Y
Product Category Fixed Effect	Y	Y	Y
Firm Fixed Effect	Y	Y	Y
Location Fixed Effect	Y	Y	Y
λ	-37.17** (2.778)	-0.0241** (0.00513)	-4.694** (0.156)
Constant	43.95** (4.597)	0.0771** (0.00847)	13.76** (0.262)
Observations	404,445	404,445	404,445

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018.

The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. ** p<0.01, * p<0.05, + p<0.1

ONLINE APPENDIX 5 – PREDICTION MODEL OF DELIVERY PRICING FOR FAVELAS

The model developed in this section explores possible predictions concerning under what conditions (when and why) consumers living inside favelas might experience higher delivery prices when purchasing products online from e-commerce firms. The development of this model is inspired by the work of (Barbosu, 2016; Chaney, 2011; Mas-Colell, Whinston, & Green, 1995; Tirole, 1988).

Consumers residing in favelas may experience problems when ordering products from e-commerce firms, mainly due to higher delivery prices. This “regressive” form of price discrimination, in which poor people are charged more, can be explained in various ways. On the one hand, the risk of losing a delivery load when entering favelas might explain higher delivery prices (Anderson et al., 2010; Duarte et al., 2019). On the other hand, there may be deliberate strategies firms use to target low-income population markets, which attenuate charging the disenfranchised more (London & Hart, 2004; Prahalad, 2006). Moreover, the consumers’ ability to hide their true type, using a zip code from outside the favela as their shipping address, might be a way to avoid extra charges. By mimicking an outside consumer zip code, favela consumers might impact the extent to which firms increase delivery prices (Mas-Colell et al., 1995).

Given this puzzle, I develop a model to explore how delivery pricing to favelas depends on the size and value (mapped into the price) of the product.⁵ The risk of robbery in a given favela and the distance from the favela resident’s home to the favela border are also considered as determinants in the pricing strategies of e-commerce firms.

SETUP

⁵ A similar reasoning can be applied to the delivery price ratio and the delivery time.

Suppose we have the following elements in the model.

Products There is a set of I different products such that each product is indexed by i , where $i \in \{1, \dots, I\}$. All products differ across two dimensions. First, the size of the product, s , where s_s is a small-sized product and s_b is a large-sized product, for example, a laptop or a book in the first case and a refrigerator or a sofa in the second case. Second, the value (price) of the products, p , where p_l is a low-value product and p_h is a high-value product. An assumption in the model is that each product i from a given e-commerce firm j has the same price for consumers living inside the favela as well as those living outside (i.e., no price discrimination of products based on residence location). This assumption is motivated by the fact that products at a given point in time are priced similarly on the e-commerce firm website, regardless of where consumers reside.

Favelas and Surrounding Areas There is a finite set of favelas, F , where $F \equiv \{1, \dots, f\}$ and a finite set of these favelas' respective surrounding areas, N , where $N \equiv \{1, \dots, n\}$. Moreover, based on the conditions inside a given favela, the risk of a robbery varies according to each favela, denoted by the probability, ρ_f where $\rho_f \in (0, 1)$. Also, I assume the probability of a robbery occurring inside a favela is at least equal than the probability of a robbery occurring outside $\rho_f \geq \rho_n \geq 0$. I assume that such a robbery will result in a loss of product i , denoted L_i .⁶ I consider $L_i \in \mathbb{R}^-$ to be the expected loss function, i.e., the probability of a robbery occurring ρ_f times the price p of the product i at e-commerce firm j , such that $L_i = E[L_i] = \rho_f \times p_{ie}$. Table 11 includes a risk L_i simulation.

⁶ For the purpose of this model, we assume that the risk of loss in favela neighboring areas is zero. However, field evidence suggest that this is not true in the reality.

Table 11 – Expected Loss Simulation (L_i)

		The Probability of Robbery (ρ_f)							
		0.1		0.25		0.5		0.75	
Price of the Product (p_{ie})	\$100	-\$ 10.00	-\$ 25.00	-\$ 50.00	-\$ 75.00	-\$ 90.00	-\$ 100.00	-\$ 100.00	-\$ 100.00
	\$500	-\$ 50.00	-\$ 125.00	-\$ 250.00	-\$ 375.00	-\$ 450.00	-\$ 500.00	-\$ 500.00	-\$ 500.00
	\$1000	-\$ 100.00	-\$ 250.00	-\$ 500.00	-\$ 750.00	-\$ 900.00	-\$ 1,000.00	-\$ 1,000.00	-\$ 1,000.00
	\$2000	-\$ 200.00	-\$ 500.00	-\$ 1,000.00	-\$ 1,500.00	-\$ 1,800.00	-\$ 2,000.00	-\$ 2,000.00	-\$ 2,000.00
	\$3000	-\$ 300.00	-\$ 750.00	-\$ 1,500.00	-\$ 2,250.00	-\$ 2,700.00	-\$ 3,000.00	-\$ 3,000.00	-\$ 3,000.00
	\$4000	-\$ 400.00	-\$ 1,000.00	-\$ 2,000.00	-\$ 3,000.00	-\$ 3,600.00	-\$ 4,000.00	-\$ 4,000.00	-\$ 4,000.00
	\$5000	-\$ 500.00	-\$ 1,250.00	-\$ 2,500.00	-\$ 3,750.00	-\$ 4,500.00	-\$ 5,000.00	-\$ 5,000.00	-\$ 5,000.00

Transportation Cost I assume that there is a transportation cost, t , to deliver a product from an e-commerce firm warehouse to the border of a given favela, which is the same for all e-commerce firms. However, I assume that due to the alleged higher risk of robbery inside a favela relative to a surrounding area, the transportation cost to a zip code inside a favela is t_f , and to a zip code outside the favela, it is t_n where $t_f > t_n > 0$. Therefore, for a given pair of a favela and its neighboring area, (f, n) , we have

$$\Delta t = t_f - t_n > 0 \quad (1)$$

Consumers I assume there are two types of consumers, c_F , and c_N , where c_F are consumers residing inside the favela, and c_N are consumers residing in surrounding areas of a favela. For the purposes of this model, the analyzes will focus only on consumers of type c_F . A consumer's utility from the consumption of product i is a function of the value that they gain from consumption and the costs incurred: (i) the price of the product, and (ii) the transportation cost of the delivery.

The utility consumer c gains from product i can be expressed as:

$$U_i^c = v_i - p_i - t_i \quad (2)$$

Favela consumers have two options, either to ask for the product to be delivered to their true zip code inside the favela (for a cost of t_f), or to a secondary zip code outside the favela (for a cost of t_n). In the case of a delivery outside the favela, I assume this takes place right at the border between the favela and a neighboring area at a reference point. There is anecdotal evidence of this phenomenon, named “*entrega no pé do morro*” (English translation: “delivery at the foot of the hill”), which was reported in interviews with informants in “*Favela da Rocinha*” (Rio de Janeiro). This process is also described in the literature on this topic (Meirelles & Athayde, 2016).⁷

In the second case, if the product is delivered at the favela border, consumers will also incur an effort cost ε to pick up the product from the outside and bring it into the favela. Let ε be a function of the product’s size, s , and the distance traveled to bring the product, d . The distance d is denoted by the distance between the consumer’s true zip code (z_i) and the zip code on the favela border, where the product was delivered (z_o). It is essential to mention that the distance distribution is skewed, many data points are quite close to the border, and few are far away, i.e. in the center of the favela area. This is because the deeper one goes to the center of a favela, the less likely a zip code is to exist, given the absence of any definition of property rights and difficulties of access (Anderson et al., 2010). Thus, there is an effort function represented by $\varepsilon(s_i, d) = s_i + d$. Table 12 includes an effort ε simulation.

Table 12 – Effort (ε) Simulation

		Distance to the Border (km)			
		0	1	2	3
Size (m ³)	book	0.03	1.03	2.03	3.03
	laptop	0.05	1.05	2.05	3.05
	sofa	1.5	2.50	3.50	4.50
	refrigerator	2	3.00	4.00	5.00

⁷ For the purpose of this model, I consider a delivery outside as a delivery on the border of the favela. Therefore, we can consider the favela neighborhood as a closed set in which the border of the favela is the set of limit points.

Therefore, the consumer participation or individual rationality (IR) constraint (i.e., for the consumer to ask for the product to be delivered to their true zip code inside the favela) is:

$$U_i^c - t_f \geq 0 \quad (\text{IR})$$

Similarly, the consumer incentive compatibility constraint (IC) is:

$$U_i^c - t_f \geq U_i^c - t_n - \varepsilon(s_i, d) \quad (\text{IC})$$

Considering a rational consumer, the IC constraint is binding, such that

$$\Delta t = \varepsilon(s_i, d) \quad (3)$$

E-commerce Firms The profit of e-commerce firms, π_e , when transacting with a consumer residing in a favela is a function of the price of product i and the transportation cost. When the product is delivered to a zip code inside the favela, the profit function can be expressed as: $\pi_e^f = (p_i - t_f - L_i)q$. This is the price of product i minus the transportation cost to deliver inside the favela, t_f , minus the expected value loss in the event of a robbery, $L_i = E[L_i] = \rho_f \times p_{ie}$, times the quantity q of the product. Similarly, the profit function when the product is delivered to a zip code on the favela border (in a surrounding area) is given by: $\pi_e^n = (p_i - t_n)q$.

I assume the market for e-commerce firms is perfectly competitive so that firms maximize profits with respect to the quantity of products sold.

$$\frac{d\pi_e^f}{dq} = \frac{d\pi_e^n}{dq} = 0$$

Moreover, as I assumed previously that each product i from e-commerce firm j has a similar price p_{ie} for consumers from both inside favelas and for those in surrounding areas at a given point in time, I find that

$$t_f - t_n = L_i$$

Consequently,

$$\Delta t = \rho_f \times p_{ie} \quad (4)$$

Predictions Combining equations (3) and (4) I show that

$$\Delta t = \varepsilon(s_i, d) = \rho_f \times p_{ie}$$

Such that:

$$\Delta t = s_i + d = \rho_f \times p_{ie} \quad (5)$$

Therefore, combining favela consumers' utility functions and e-commerce firms' profit maximization functions, the model generates four predictions:

Prediction 1: The higher the transaction cost, measured by the alleged higher risk of robbery inside a favela ρ_f , the higher the e-commerce firm's delivery pricing, Δt .

Prediction 2: The greater the distance d between favela consumers' true zip code and the favela border, the higher the e-commerce firm's delivery pricing, Δt .

Prediction 3: The higher the value (price) p of product i at e-commerce firm j , the higher the e-commerce firm's delivery pricing to favela consumers, Δt .

Prediction 4: The larger the size s of product i at e-commerce firm j , the higher the e-commerce firm's delivery pricing to favela consumers, Δt .

ONLINE APPENDIX 6 – E-COMMERCE FIRMS CHARACTERISTICS AND COMPARISON

Id Firm	Founding Year	Firm has Physical Stores (dummy)	Total Number of Stores (count)	Firm has Physical Stores in Favelas (dummy)	Employees' Assessment of the Firm Culture (score)	Responsiveness to Consumer Complaints (reply rate)	Estimated Market Share (in %)	Estimated Revenue (in U\$ millions)
Firm #01	1929	Yes	1,320	No	3.7	99.4%	8.3%	U\$ 1,051
Firm #02	1959	Yes	570	No	3.7	98.6%	1.4%	U\$ 427
Firm #03	1952	Yes	772	Yes	4.0	99.7%	3.0%	U\$ 716
Firm #04	1989	Yes	640	No	3.8	99.8%	3.1%	U\$ 594
Firm #05	2003	No	0	No	3.8	99.8%	0.5%	N/A
Firm #06	1980	Yes	140	No	3.5	99.1%	0.5%	N/A
Firm #07	1946	Yes	222	No	3.9	99.7%	4.5%	U\$ 206
Firm #08	1989	Yes	650	Yes	3.5	99.2%	0.7%	N/A
Firm #09	1914	Yes	100	No	4.0	100.0%	1.1%	U\$ 57
Firm #10	1999	No	0	No	3.7	99.6%	4.1%	U\$ 252
Firm #11	1962	Yes	450	No	3.4	100.0%	2.3%	U\$ 138

Source: Euromonitor, Indeed.com, *Reclame Aqui*, SBVC (2016, 2020), and e-commerce firms websites. Note: estimated revenue for public traded firms considers the percent of revenue from each firm's online business unit in 2019, data from SBVC (2020) converted to U\$ dollars. Euromonitor estimates that the Brazilian internet retailing market was R\$ 75 billion [U\$ 18 billion] in 2019, whereas consumer to consumer (C2C) transactions represented from 30% to 40% of the total e-commerce market share.

#	Variable	Obs.	Min	Max	Mean	Std. Dev.	Correlation Matrix					
							1	2	3	4	5	6
1	Firm has Physical Stores (dummy)	11	0	1	0.82	0.40	1.000					
2	Total Number of Stores (count)	11	0	1,320	442	403	0.542+	1.000				
3	Firm has Physical Stores in Favelas (dummy)	11	0	1	0.18	0.40	0.222	0.330	1.000			
4	Employees' Assessment of the Firm Culture (score)	11	3.4	4.0	3.7	0.2	-0.056	-0.069	0.056	1.000		
5	Responsiveness to Consumer Complaints (reply rate)	11	98.6%	100.0%	99.5%	0.4%	-0.189	-0.236	-0.100	0.338	1.000	
6	Estimated Market Share (in %)	11	0.5%	8.3%	2.7%	2.3%	0.081	0.601+	-0.176	0.161	0.112	1.000
7	Estimated Revenue (in U\$ millions)	8	U\$ 57	U\$ 1,051	U\$ 430	U\$ 338	0.213	0.921**	0.342	0.070	-0.326	0.684+

Note: ** p<0.01, * p<0.05, + p<0.1

The correlation table comparing the characteristics of firms in the sample shows a strong positive correlation (correl. 0.921) between a firm's revenue and the total number of stores they have ($p < 0.01$). As expected, there is also a positive correlation between revenue and market share ($p < 0.1$). Interestingly, there is no significant correlation between revenue and the two measures of stakeholder orientation. Overall, this analysis points out that firms with larger revenue may be more likely to invest in physical stores. However, the stakeholder orientation of firms is not related to level of revenue. This conclusion is not conclusive because there is revenue information for only eight out of eleven firms in the sample. I also do not have information on firm profitability inside and outside of favelas.

Finally, the paper presents a categorization of firms accordingly to their situation in resources and capabilities and stakeholder management. This classification enables a systematic comparison among firms that combine both resources and stakeholder orientation when dealing with favela's consumers.

		Stakeholder Management (Employee's Assessment of the Firm Culture)	
		<i>Below or Equal Median</i>	<i>Above Median</i>
Resources and Capabilities (Physical Store)	<i>Online Only</i>	Firm #10	Firm #05
	<i>Outside Favelas</i>	Firms #01, #02, #06, and #11	Firms #04, #07, and #09
	<i>Outside and Inside Favelas</i>	Firm #08	Firm #03

Note: This classification is merely illustrative for the purpose of the theory integration test.

ONLINE APPENDIX 7 – VISUAL EVIDENCE FROM THE FIELD



Note: Favela entry point (top) and e-commerce firm physical store in favela da rocinha (bottom). Pictures taken by the author. The e-commerce firm #03's logo was covered, aiming to maintain anonymity. Similarly, individuals' faces were blurred (de-identified) to respect their privacy.

ONLINE APPENDIX 8 – SLOPE SIGNIFICANCE ANALYSIS

I performed additional analysis to evaluate whether the distance curve is significant for outside and inside favelas zip codes. These analyses help to double-check whether the slope of the distance curve (presented in Figure 4 of the main paper) is indeed significant. Table 13 shows the result for ***outside favela distances***. The slope is not significant in any of the measures, both main and alternative dependent variables.

Table 14 presents the same test, but for ***inside favela distances***. In this case, the slope is not significant for “Refused Order” and its alternative measure, “Out of Stock.” However, for delivery price, delivery price ratio, and delivery time variables, we have a significant slope ($p < 0.01$), which the literature describes as a “regression kink design” (Nielsen, Sørensen, & Taber, 2010). These results validate the conclusion that the inner a zip code is, the more e-commerce firms charge to favela consumers.

Table 13 – Distances from Outside Favela

	LOGIT (1) 1(Refused Order)	LOGIT (2) 1(Out of Stock)
Distance (outside favela)	-0.00102 (0.0793)	0.0228 (0.0755)
Constant	-1.990** (0.0825)	-2.130** (0.0849)
Observations	212,765	212,765
Pseudo R-squared	0.000	0.000
Log-Likelihood	-78213	-71649

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018.

The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. ** p<0.01, * p<0.05, + p<0.1

	OLS (1) Delivery Price (in R\$)	OLS (2) Delivery Price Ratio (in %)	OLS (3) Delivery Time (in Days)
Distance (outside favela)	-13.63 (12.71)	-0.0313 (0.0260)	-0.691 (0.890)
Constant	74.58** (13.45)	0.217** (0.0290)	15.39** (0.967)
Observations	185,789	185,789	185,789
R-squared	0.001	0.001	0.000

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018.

The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. ** p<0.01, * p<0.05, + p<0.1

Table 14 – Distances from Inside Favela

	LOGIT (1) 1(Refused Order)	LOGIT (2) 1(Out of Stock)
Distance (inside favela)	-0.197 (0.199)	-0.264 (0.200)
Constant	-1.865** (0.141)	-2.002** (0.147)
Observations	194,135	194,135
Pseudo R-squared	0.000458	0.000784
Log-Likelihood	-74072	-67691

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018.

The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. ** p<0.01, * p<0.05, + p<0.1

	OLS (1) Delivery Price (in R\$)	OLS (2) Delivery Price Ratio (in %)	OLS (3) Delivery Time (in Days)
Distance (inside favela)	96.28** (19.84)	0.213** (0.0473)	6.726** (1.555)
Constant	65.22** (12.88)	0.192** (0.0271)	14.44** (0.823)
Observations	168,288	168,288	168,288
R-squared	0.024	0.032	0.033

Note: standard errors are clustered at twenty-one different favela-surrounding dyads. The data collection occurred once a week throughout seventeen weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018.

The data has information on ten different products of four different categories (by size and value), for eleven different e-commerce firms, operating in the main state capitals of Brazil. ** p<0.01, * p<0.05, + p<0.1

ONLINE APPENDIX 9 – MCCRARY TEST AND BANDWIDTH ANALYSIS

To verify the robustness of the regression discontinuity design to the present setting, I performed additional analyses varying the bandwidths. The bandwidths' selection was defined based on the zip code distance to the favela border distribution, as shown in Figure 5. Given this distribution, I performed tests ranging from a 750m bandwidth to a 50m bandwidth. Table 15 presents the results of these analysis on the paper's main dependent variables.

Also, following the RDD literature (Flammer, 2015; Jacob & Zhu, 2012; McCrary, 2008), Figure 6 shows a McCrary (2008) test to check the smoothness of the density function at the threshold (the favela border). The test does not reject the null hypothesis of the discontinuity in the density function at the cut-off point being equal to zero (p-value = 0.690).

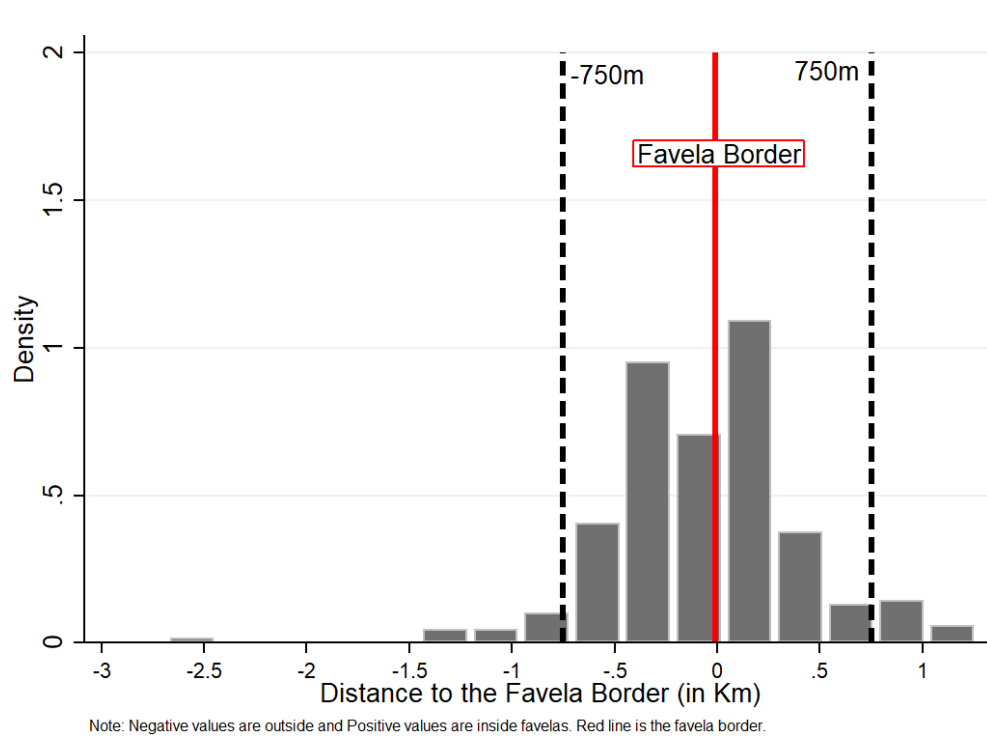


Figure 5 – Distribution of Zip Codes by Distances to the Favela Border (red line)

Note: Negative values are outside and positive values are inside favelas. The Red line represents the favela border.

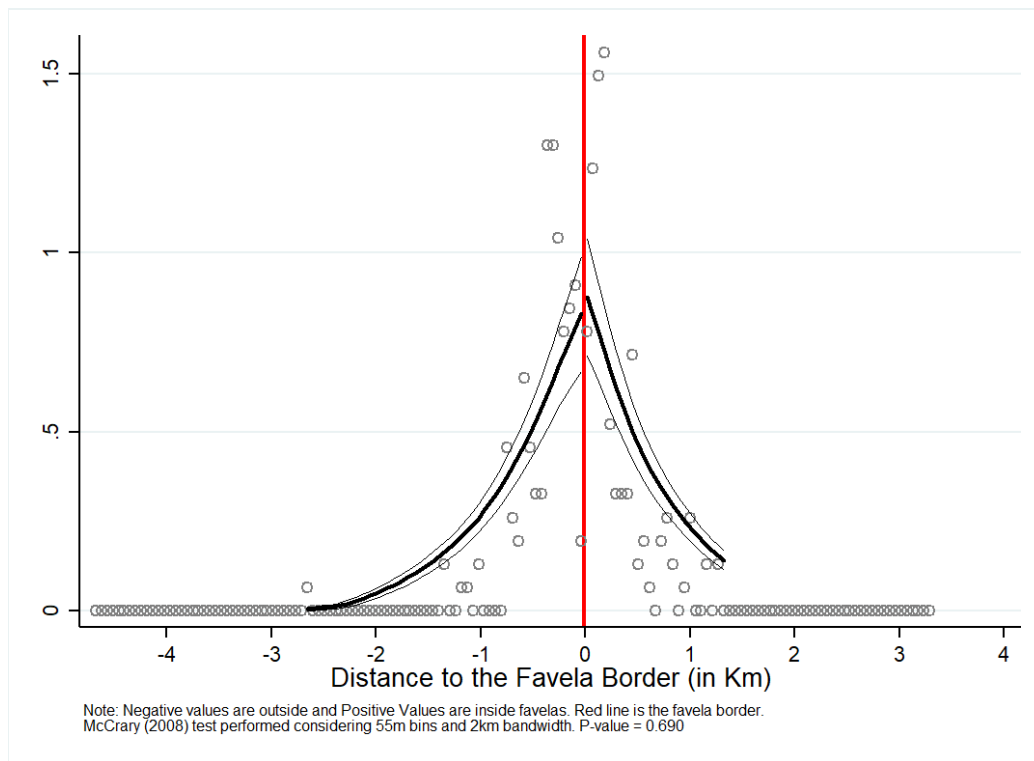


Figure 6 – Visualization of the McCrary (2008) Test

Note: Negative values are outside and positive values are inside favelas. The red line represents the favela border. McCrary (2008) test was performed considering 55 meters bins and 2 kilometers bandwidth. The decision for 2 km bandwidth was subjectively defined and motivated by the fact that the full sample distance window is already reasonably narrow (-2.68km to 1.27km). McCrary (2008) test's p-value = 0.690.

Table 15 – RDD Bandwidth Analyses on main Dependent Variables (Part 1/3)

	LOGIT (1)	LOGIT (2)	LOGIT (3)	LOGIT (4)	LOGIT (5)	LOGIT (6)
			1(Refused Order)			
Bandwidths	Full Sample	750m	500m	250m	100m	50m
Favela	0.145** (0.00191)	0.156* (0.0674)	0.180* (0.0717)	0.209* (0.0956)	0.131 (0.132)	0.0454** (0.00107)
Week Fixed Effect	Y	Y	Y	Y	Y	Y
Product Category Fixed Effect	Y	Y	Y	Y	Y	Y
Firm Fixed Effect	Y	Y	Y	Y	Y	Y
Location Fixed Effect	Y	Y	Y	Y	Y	Y
Constant	0.513** (0.114)	1.149** (0.142)	1.154** (0.149)	1.167** (0.184)	1.460** (0.205)	1.710** (0.174)
Observations	288,679	261,325	224,859	137,750	41,734	13,949
Pseudo R-Squared	0.147	0.147	0.149	0.147	0.145	0.154
Log-likelihood	-114,032	-103,364	-88,991	-55,891	-16,928	-5,914

Note: standard errors are clustered at 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (size, value), for eleven different e-commerce firms, operating in 21 different favelas and neighborhoods at the main state capitals in Brazil. ** p<0.01, * p<0.05, + p<0.1

Table 15 – RDD Bandwidth Analyses on main Dependent Variables (Part 2/3)

	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)
	Delivery Price (in Brazilian Reais - R\$)					
Bandwidths	Full Sample	750m	500m	250m	100m	50m
Favela	18.65** (0.0937)	47.72** (0.650)	47.79** (0.648)	47.87** (0.675)	48.24** (0.797)	9.924** (0.787)
Week Fixed Effect	Y	Y	Y	Y	Y	Y
Product Category Fixed Effect	Y	Y	Y	Y	Y	Y
Firm Fixed Effect	Y	Y	Y	Y	Y	Y
Location Fixed Effect	Y	Y	Y	Y	Y	Y
Constant	19.17* (9.001)	-6.988 (8.212)	-6.158 (7.873)	-3.386 (7.651)	-0.564 (8.725)	-0.0376 (9.987)
Observations	354,077	320,319	275,377	167,630	51,911	17,099
R-squared	0.334	0.328	0.326	0.318	0.308	0.300

Note: standard errors are clustered at 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (size, value), for eleven different e-commerce firms, operating in 21 different favelas and neighborhoods at the main state capitals in Brazil. ** p<0.01, * p<0.05, + p<0.1

Table 15 – RDD Bandwidth Analyses on main Dependent Variables (Part 3/3)

	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)
	Ratio Delivery Price over Price of the Product (in %)					
Bandwidths	Full Sample	750m	500m	250m	100m	50m
Favela	0.0306** (0.000169)	0.0864** (0.000710)	0.0865** (0.000722)	0.0868** (0.000781)	0.0856** (0.000890)	0.0323** (0.000356)
Week Fixed Effect	Y	Y	Y	Y	Y	Y
Product Category Fixed Effect	Y	Y	Y	Y	Y	Y
Firm Fixed Effect	Y	Y	Y	Y	Y	Y
Location Fixed Effect	Y	Y	Y	Y	Y	Y
Constant	0.0610** (0.0184)	0.0118 (0.0168)	0.0138 (0.0158)	0.0173 (0.0147)	0.0316 (0.0208)	0.0136 (0.0225)
Observations	354,077	320,319	275,377	167,630	51,911	17,099
R-squared	0.385	0.385	0.384	0.381	0.381	0.367

Note: standard errors are clustered at 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (size, value), for eleven different e-commerce firms, operating in 21 different favelas and neighborhoods at the main state capitals in Brazil. ** p<0.01, * p<0.05, + p<0.1

ONLINE APPENDIX 10 – INFRASTRUCTURE ANALYSIS

Below, the study presents a test to affirm that the order refusals and higher delivery prices charged were identified by the locations of the customer zip codes rather than by potential differences in physical infrastructure between zip codes that lie marginally inside vs. marginally outside the favela boundaries. To do so, I use information on three infrastructure variables: (i) **electricity coverage** (in %); (ii) **garbage collection coverage** (in %); and (iii) **water and sewage coverage** (in %) for each zip code in the dataset. Table 16 and Figure 7 shows that, for all three measures, discontinuities do not arise at the threshold of the favela border. Also, only for water and sewage (which is usually underground) variable there statistical difference ($p < 0.01$) the further inside favela a zip code is located. This suggests that the logistical challenges of delivering products to zip codes located marginally inside vs. marginally outside the favelas border are roughly equivalent, which helps validating the RDD assumptions.⁸

Table 16 - Discontinuity Test on Infrastructure Coverage

	OLS (1) Electricity Coverage (in %)	OLS (2) Garbage Collection Coverage (in %)	OLS (3) Water and Sewage Coverage (in %)
Favela	-0.00526 (0.00361)	-0.0153 (0.00890)	-0.0366 (0.0248)
Distance (in Km)	-0.00289 (0.00322)	-0.0168+ (0.00839)	-0.0733** (0.0170)
Location Fixed Effect	Y	Y	Y
Constant	1.002** (0.00164)	0.880** (0.00503)	0.958** (0.0122)
Observations	406,900	406,900	406,900
R-squared	0.551	0.650	0.723

Note: standard errors are clustered at 21 different favela-surrounding dyads. ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

⁸ To provide a vivid illustration of the RDD assumption test on logistical infrastructure inside favelas, please check the Washington Post's 360° video report available at https://www.youtube.com/watch?v=x_35zVvJtAg&feature=youtu.be. Note on 0'59'' minute a garbage collection truck in operation inside the favela. Accessed on September 13th, 2020.

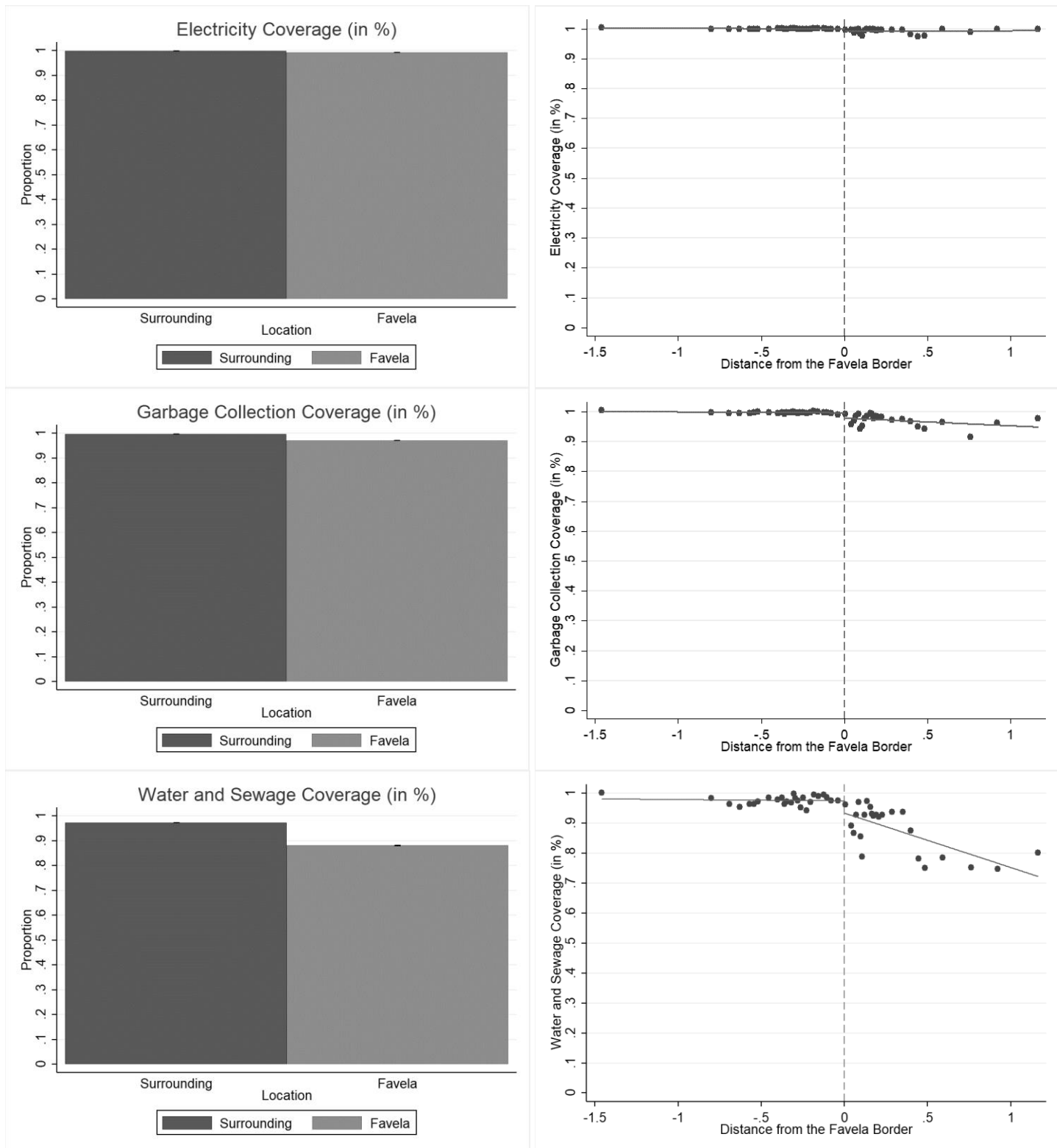


Figure 7 – Discontinuity Test on Infrastructure Coverage

Source: Brazilian National 2010 Census Data collected at the *Atlas do Desenvolvimento Humano no Brasil*.

Note: The estimation was performed considering 50 bins and a discontinuity point at the favela border. All negative distances represent zip codes outside the favelas and all positive distances zip codes inside the favelas. Graphs y-axes ranging from 0% to 100%.

ONLINE APPENDIX 11 – CRIMINALITY ANALYSIS

A reasonable concern for deliveries to favelas is the potential higher criminality risk of these communities. Unfortunately, there are not specific criminality indicators by zip code in Brazil; therefore, I cannot entirely dismiss this hypothesis. However, aiming to provide a suggestive indication that criminality is not the primary driver behind firms' behavior, the paper leveraged data with the *Instituto de Segurança Pública* (en. Public Security Institute of Rio de Janeiro State – ISP) and presents additional analyses on criminality comparing robbery indicators from favelas with *Unidades de Polícia Pacificadora* (en. Pacifying Police Units - UPPs) and the average of the Rio de Janeiro metropolitan area.

The UPP project started in 2007 and was design to reassure the control of favelas by the state in the upcoming 2014 FIFA World Cup and the 2016 Summer Olympic Games (Bellego & Drouard, 2019; Jovchelovitch & Priego-Hernandez, 2013; Larkins, 2015; Willis & Prado, 2014). The program implemented police stations inside 38 different favelas of the Rio city and collected systematic data on crime within these areas. I use this data to compare crime occurrences, particularly robbery/theft within favelas vis-à-vis the whole Rio metropolitan area. This comparison aims to prove that crime rates are not systematically different between favelas and outside favela areas.

Results presented in Table 17 and visually illustrated in Figure 8 reinforce this interpretation. Defining the countryside of Rio de Janeiro State as the baseline, the results show that crime inside favelas is not necessarily higher than in the Rio metropolitan area. Concerning e-commerce firms' delivery strategies, two types of robberies are of particular interest: **cargo robbery** (Table 17, Column 5) and **store robbery** (Table 17, Column 6). On the one hand, results show that favelas have, on average, 1.7 more cargo robberies than countryside locations ($p < 0.01$);

however, there is no statistical difference in store robbery. On the other hand, the whole metropolitan area has, on average, 7.0 more cargo robberies than countryside locations. Despite the marginal significance of this result ($p < 0.1$), the magnitude difference is uncanny.

Moreover, reinforcing some of the qualitative findings, the average of store robberies is higher in whole metropolitan areas – 2.1 more cases ($p < 0.1$) – than in favelas. Once more, the result is marginally significant but informative. The overall analysis suggests that criminality can be a concern for e-commerce delivery. However, crime rates inside favelas do not seem to be much different from the average of the whole metropolitan area.

These analyses have remarkable limitations that should be accounted for. First, the sample of 38 favelas with UPP are located only in Rio de Janeiro. Different from the main paper, which has favelas across various cities and states in Brazil. Second, despite some overlap with the favelas in this paper's main sample – such as favelas as *Rocinha*, *Complexo do Alemão*, and *Jacarezinho* – many favelas are uniquely identified in the ISP UPP database.⁹ Third, potential reporting biases are expected since favela populations might prefer not to report a crime to the police, so there might be selection and under-representation of reported crimes in UPPs. Finally, the models do not include controls for population density as this data is not available monthly. Were this data available, I believe its inclusion would reinforce the findings because favelas typically have a higher population density than adjacent urban areas, as described in APPENDIX 1.

⁹ Additional analyses considering only the four favelas that are in both samples were performed. The results are robust to this specification. I opted to include the full sample, aiming to provide the most generalizable picture possible.

Table 17 – Comparative Analysis on Robbery incidents inside Favelas and in the whole Rio Metropolitan Area

	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)	OLS (7)	OLS (8)
	Robbery by Type (Number of Occurrences per Month)							
	Street Robbery	Cellphone Robbery	Bicycle Robbery	Automobile Robbery	Cargo Robbery	Store Robbery	Home Robbery	All Robberies
Favelas with UPPs	14.61** (2.820)	3.002** (0.466)	0.0164** (0.00463)	6.019** (1.130)	1.723** (0.271)	0.178 (0.120)	-0.160** (0.0349)	29.57** (4.671)
Rio Metropolitan Area	63.10** (22.83)	6.325* (2.748)	-0.0937+ (0.0548)	47.55** (15.19)	7.001+ (3.742)	2.137+ (1.089)	0.631** (0.239)	117.9** (41.68)
Controls for the								
Number of Police Reports	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y	Y	Y	Y
Constant	-40.98** (11.38)	-14.07* (5.444)	-0.119+ (0.0682)	-12.57** (3.443)	-3.342** (0.744)	0.211 (0.307)	0.199** (0.0581)	-69.33** (19.67)
Observations	9,360	9,360	9,360	9,360	9,360	9,360	9,360	9,360
R-squared	0.965	0.864	0.602	0.911	0.871	0.962	0.897	0.957

Note: standard errors are clustered at 130 different locations, 38 different favelas, and 92 different municipalities in the Rio de Janeiro state. The data ranges from January 2014 to December 2019. The baseline of the estimations is Rio de Janeiro state countryside municipalities. ** p<0.01, * p<0.05, + p<0.1

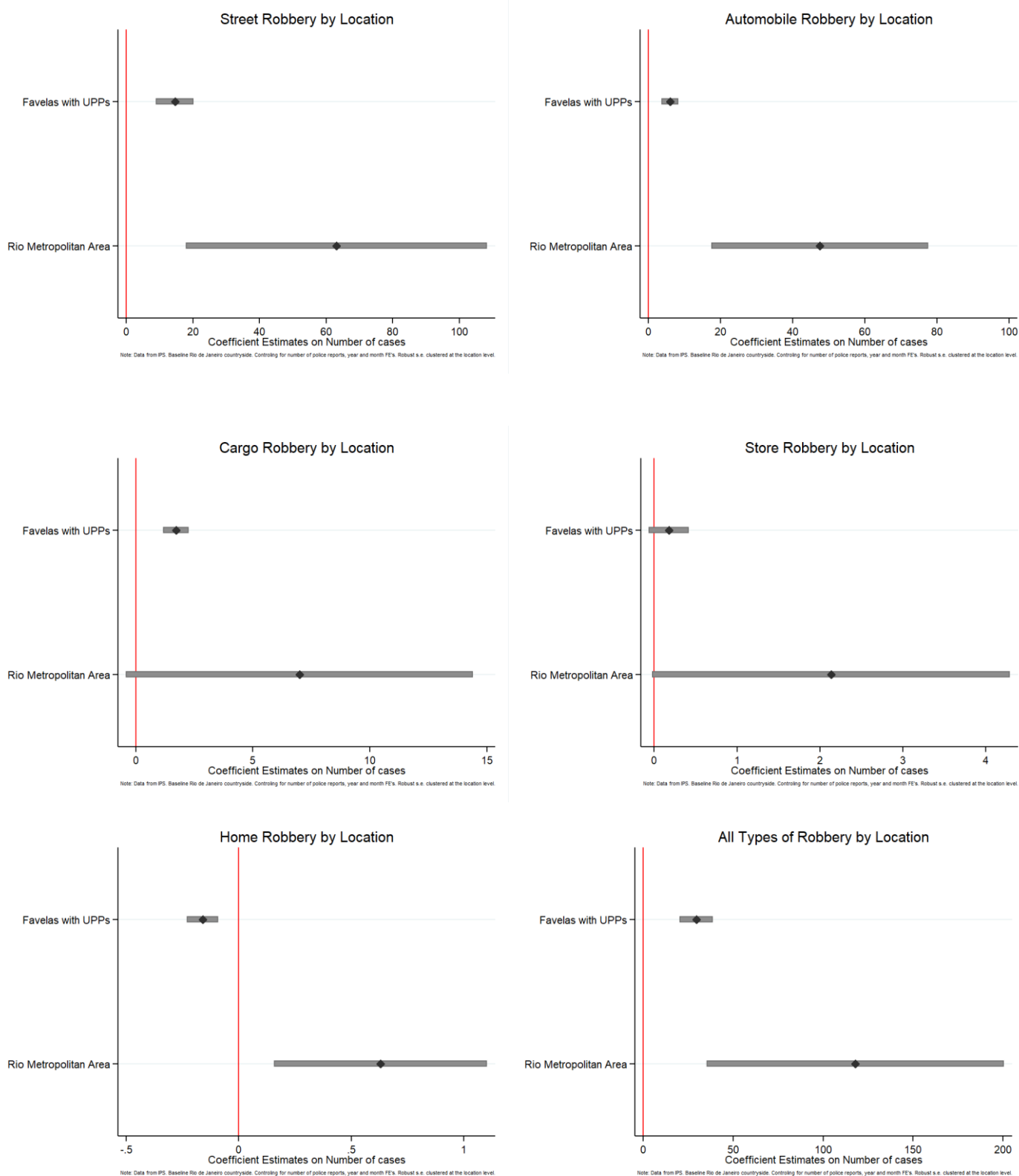


Figure 8 – Coefficient Plots on Robbery Incidents inside Favelas and Rio Metropolitan Area

Note: standard errors are clustered at 130 different locations, 38 different favelas, and 92 different municipalities in the Rio de Janeiro state. The data ranges from January 2014 to December 2019. The baseline of the estimations is Rio de Janeiro state countryside municipalities. The graphs show the point estimate and 95% control interval. Graphs x-axes in different scales.

ONLINE APPENDIX 12 – OLS TEST ON THE BINARY OUTCOMES

A robustness check on the binary dependent variables – **1(Refused Orders)** and the alternative binary dependent variable **1(Out of Stock)** – is presented in Table 18. This analysis also considers firms for which these dependent variables have never changed throughout the whole period of analysis. The results are significant ($p < 0.01$) and point in the same direction that the main Logit models, i.e. favelas have 1.13 p.p. higher likelihood to have their orders refused and 0.21 p.p. higher chance to receive an “out of stock” message.

Table 18 – Robustness Checks using OLS on Binary Outcomes, Main and Alternative

	OLS (1) 1(Refused Order)	OLS (2) 1(Out of Stock)
Favela	0.0113** (8.73e-06)	0.00213** (8.29e-06)
Week Fixed Effect	Y	Y
Product Category Fixed Effect	Y	Y
Firm Fixed Effect	Y	Y
Location Fixed Effect	Y	Y
Constant	0.483** (0.00727)	0.486** (0.00714)
Observations	406,900	406,900
R-squared	0.164	0.197

Note: standard errors are clustered at 21 different favela-surrounding dyads. The data collection occurred once a week throughout 17 weeks, usually on Sundays from October 15th, 2017 to February 17th, 2018. The data has information on ten different products of four different categories (size, value), for eleven different e-commerce firms, operating in 21 different favelas and neighborhoods at the main state capitals in Brazil. ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

ONLINE APPENDIX 13 – FAVELA CONSUMERS’ PURCHASE AND PURCHASE CHANNELS

Additional information on favela consumers' purchases and purchase channel preferences are described below. This data was surveyed and collected by *Instituto Data Favela and Locomotiva*, and publicized in *Valor Econômico*, a renowned economic analysis newspaper in Brazil, on January 24th, 2020 under the title (in English): “*Favelas have the purchasing power of R\$ 119.8 billion*¹⁰” (Villas Bôas, 2020).

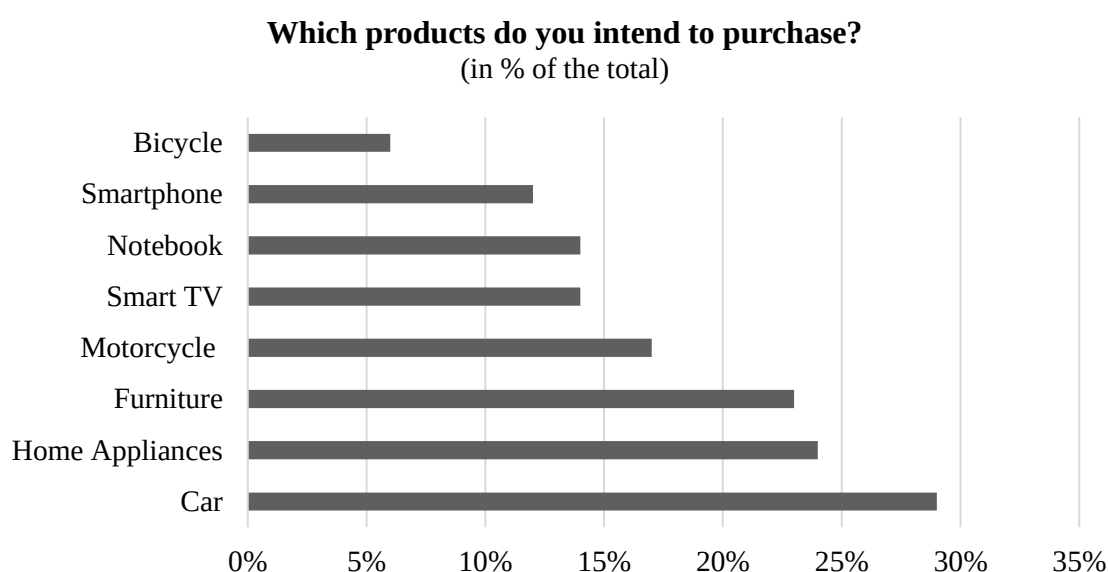


Figure 9 – Durable Goods Purchase Intension by Favela Consumers

Note: data from Instituto Data Favela and Locomotiva, available at Valor Econômico (Villas Bôas, 2020).

The report is based on 2.670 interviews in 465 favelas in Brazil. Figure 9 provides a picture of which products favela residents intended to purchase, and Table 19 describes by which purchase channel. Although there is a higher preference for physical stores, online or hybrid (physical stores and online) channels can range from 20% to almost 50% of preference, depending on the product.

¹⁰ This value equals to U\$ 28.7 billion in January 24th, 2020 – the date this report was published.

The same report describes that more than one-third (36%) of the favela interviewees declared not receiving their internet purchases at home (Villas Bôas, 2020), which endogenously might affect their responses about preferred purchase channels.

Table 19 – Sales Channel Preference by Favela Consumers

Through which channel do you plan to perform the purchase? (in % of the total)							
	Bicycle	Smartphone	Notebook	Smart TV	Motorcycle	Furniture	Home Appliances
Physical Stores	66%	52%	43%	59%	71%	72%	67%
Online	6%	7%	11%	7%	1%	4%	6%
Both/Either	15%	32%	37%	26%	19%	17%	20%
Not sure	13%	10%	8%	9%	9%	8%	7%

Note: data from Instituto Data Favela and Locomotiva, available at Valor Econômico (Villas Bôas, 2020).

ONLINE APPENDIX 14 – DISTANCE DATA COLLECTION ILLUSTRATION



Figure 10 – Distances between Zip Codes and the Favela Border, *Favela da Rocinha*

Source: Google Earth® software.

The screen-shots describe how the distance data was collected. In both pictures, we can see Favela da Rocinha (pink shaded area) and its surroundings (non-shaded areas). The yellow line measures the linear distance between a zip code and the main entry point of the favela. In the left picture, we have an illustration for a zip-code outside the favela (in São Conrado neighborhood) and the favela main entry point at its southern border. Whereas, in the right picture, we have an illustration of the distance between a zip code inside the favela and the favela main entry point at its southern border. An on-site image of this entry point can be found in the Online Appendix 7.

To be conservative, I manually collected the distance data considering two factors: (a) the location of the main entry points and (b) the closest possible main entry point to each zip code. Finally, to operationalize the econometric models, distances d were registered with negative values for zip codes *outside* favelas $d_z^{Surrounding} < 0$ and with positive values for zip codes *inside* favelas $d_z^{Favela} > 0$.

ONLINE APPENDIX REFERENCES

- Anderson, J., Markides, C., & Kupp, M. (2010). The Last Frontier: market creation in conflict zones, deep rural areas, and urban slums. *California Management Review*, 52(4), 6–28. <https://doi.org/10.1525/cmr.2010.52.4.6>
- Barbosu, S. (2016). *Uncovering Latent Clusters and Consumer Preferences Leading to their Formation*.
- Bellego, C., & Drouard, J. (2019). Does It Pay to Fight Crime? Evidence From the Pacification of Slums in Rio de Janeiro. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3326563>
- Chaney, T. (2011). *The Gravity Equation in International Trade: an Explanation*. Retrieved from <http://www.jstor.org/stable/40441082%5Cnhttp://about.jstor.org/terms>
- Duarte, A. L. de C. M., Macau, F., Flores e Silva, C., & Sanches, L. M. (2019). Last mile delivery to the bottom of the pyramid in Brazilian slums. *International Journal of Physical Distribution & Logistics Management*, IJPDLM – 01–2018 – 0008. <https://doi.org/10.1108/IJPDLM-01-2018-0008>
- Flammer, C. (2015). Does Corporate Social Responsibility Lead to Superior Financial Performance ? A Regression Discontinuity Approach. *Management Science*, 61(11), 2549–2568.
- Goldfarb, A., & Tucker, C. (2017). Inequality, privacy, and digital market design. In S. Kominers & A. Teytelboym (Eds.), *Fair by Design*. Oxford University Press.
- Heckman, J. (1979). Sample Specification Bias as a Selection Error. *Econometrica*, 47(1), 153–162.
- Jacob, R., & Zhu, P. (2012). *A practical guide to Regression Discontinuity*. MDRC. https://doi.org/10.1007/978-1-4302-4633-6_11
- Jovchelovitch, S., & Priego-Hernandez, J. (2013). *Underground Sociabilities: Identity, Culture and Resistance in Rio de Janeiro's Favelas*. Brasilia: UNESCO.
- Larkins, E. R. (2015). *The Spectacular Favela: Violence in Modern Brazil*. University of California Press.
- London, T., & Hart, S. L. (2004). Reinventing Strategies for Emerging Markets: Beyond the Transnational Model. *Journal of International Business Studies*, 35(5), 350–370. <https://doi.org/10.1057/palgrave.jibs.8400099>
- Mas-Colell, A., Whinston, M. D., & Green, J. R. (1995). *Microeconomic Theory* (First). New York: Oxford University Press.
- McCrary, J. (2008). Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics*, 142(2), 698–714. <https://doi.org/10.1016/j.jeconom.2007.05.005>
- Meirelles, R., & Athayde, C. (2016). *Um país chamado favela*. Editora Gente Liv and Edit Ltd.
- Nielsen, H. S., Sørensen, T., & Taber, C. (2010). Estimating the effect of student aid on college enrollment: Evidence from a government grant policy reform. *American Economic Journal: Economic Policy*, 2(2), 185–215. <https://doi.org/10.1257/pol.2.2.185>
- North, D. C. (1992). *Transaction Costs, Institutions, and Economic Performance*. San Francisco, CA: ICEG Press. <https://doi.org/10.1111/j.1465-7295.1987.tb00750.x>
- Perlman, J. (2010). *Favela: Four Decades of Living on the Edge in Rio de Janeiro*. New York: Oxford University Press.
- Prahalad, C. K. (2006). *Fortune at the Bottom of the Pyramid: Eradicating Poverty through Profits*. Philadelphia, PA.: Wharton School Publishing.
- Puhani, P. A. (2000). The Heckman correction for sample selection and its critique. *Journal of*

- Economic Surveys*, 14(1), 53–68. <https://doi.org/10.1111/1467-6419.00104>
- SBVC. (2016). *Ranking SBVC 50 Maiored Empresas E-commerce Brasileiro 2016*. São Paulo, Brazil.
- SBVC. (2020). *300 Maiores Empresas do Varejo Brasileiro*. Sociedade Brasileira de Varejo e Consumo.
- Tirole, J. (1988). *The Theory of Industrial Organization*. MIT Press.
- Villas Bôas, B. (2020, January 24). Favelas têm poder de compra de R\$ 119,8bi. *Valor Econômico*. Retrieved from <https://valor.globo.com/brasil/noticia/2020/01/24/favelas-tem-poder-de-compra-de-r-1198-bi.ghtml>
- Willis, G. D., & Prado, M. M. (2014). Process and Pattern in Institutional Reforms: A Case Study of the Police Pacifying Units (UPPs) in Brazil. *World Development*, 64, 232–242. <https://doi.org/10.1016/j.worlddev.2014.06.006>